

Cross-Country Differences in the Impacts of Sectoral Shocks*

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Abstract

The same sectoral shock can devastate a poor country and leave a rich one largely unaffected. I explain this contrast in a multisector model with input–output linkages and non-homothetic preferences, in which GDP and welfare respond differently to identical shocks. Calibrating the model to 80 economies, I find that a common agricultural shock produces median welfare losses 19.0 times larger in the poorest quartile of countries than in the richest, against a factor of 1.46 for GDP. The vulnerability of poor countries stems mainly from subsistence demand, and the role of production networks itself changes as countries develop.

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1 Introduction

The impact of sectoral shocks on the economy differs sharply between rich and poor countries, and the contrast is particularly striking in agriculture. When an adverse shock to agriculture, such as extreme heat, severe drought, or flooding, strikes a poor country, aggregate output falls markedly, with substantial consequences for household consumption and living standards. In rich countries, comparable shocks to agriculture tend to have much smaller effects on aggregate output and average household well-being.¹

The conventional way to interpret this pattern runs through Hulten’s theorem: in efficient economies, an industry’s sales as a share of GDP, its Domar weight, is a sufficient statistic for the first-order impact of a sectoral shock on aggregate output (Hulten, 1978). Through this lens, the pattern seems easy to explain. Agriculture looms large in poor economies’ sales: in the data of Section 5, the observed Domar weight of agriculture has a median of 0.406 in the poorest quarter of the countries studied, against 0.12 in the richest. Bigger weight, bigger exposure: the explanation seems complete. The point of this paper is that this explanation is incomplete when preferences are non-homothetic: *Hulten’s theorem then fails with respect to GDP but holds for welfare measures up to a country-specific scalar*. As a result, the observed Domar weight can overstate or understate how much GDP and welfare respond to such shocks.²

I make this point in two steps, which are the paper’s two main contributions. The first is theoretical. In a canonical multisector model with input–output linkages and non-homothetic preferences, I characterize in closed form how sectoral shocks affect GDP, how they affect welfare, and what the observed sales shares can and cannot say about either. The three answers separate where non-homotheticity matters. The second is quantitative. Calibrating the model to 80 economies, I find that the same agricultural shock costs poor countries an order of magnitude more in welfare than in GDP, while in rich countries the two costs nearly coincide; that the observed sales shares accordingly mismeasure the responses most severely in poor countries; and that subsistence demand determines which countries are vulnerable, while the role of production networks transforms along the development path, from amplifying subsistence needs in poor countries to transmitting costs in rich ones.

The first part of the paper concerns the theory. To analyze the differences in the impacts

¹See Da-Rocha and Restuccia (2006) on agriculture and cross-country output volatility, and Dell, Jones, and Olken (2012), Zappalà (2023), and Akyapı, Bellon, and Massetti (2025) on the output effects of adverse weather shocks. Household-level evidence documents limited consumption smoothing during droughts and sizable consumption losses from rainfall shortfalls, with adverse consequences for living standards (Kazianga & Udry, 2006; Hill & Porter, 2017).

²The cross-country differences have many plausible contributors besides the mechanism studied here: technological diversification, financial development, labor mobility, land institutions, and obstacles to trade, among others (see, e.g., Koren & Tenreyro, 2013; Banerjee & Duflo, 2005; Donovan, Lu, & Schoellman, 2023; Pande & Udry, 2006; Atkin & Donaldson, 2022). The paper’s claim is not that these forces are absent, but that the mechanism it isolates, the interaction of production networks and non-homothetic preferences, is quantitatively important on its own.

of sectoral shocks between rich and poor countries, I consider a canonical production-network model of [Acemoglu, Carvalho, Ozdaglar, and Tahbaz-Salehi \(2012\)](#) with one deviation: the final-demand aggregator is non-homothetic.³ The model admits a closed-form solution. GDP, measured with a fixed-weight deflator, responds to sectoral shocks with the Domar weights of the economy’s homothetic benchmark, not with its own observed Domar weights. Welfare responds with the observed Domar weight, amplified by the inverse of the discretionary share: one divided by the share of income left over after committed consumption. The observed Domar weight, the practitioner’s sufficient statistic, can therefore be a poor proxy for both responses at once. In the leading case in which food is a single necessity, the directions are sharp: it overstates the GDP impact of agricultural shocks, because the committed part of food sales adds to the Domar weight but not to the GDP response, and it understates their welfare impact, because the same committed position is what makes each additional unit of income so valuable near subsistence. Example economies, with different forms of non-homotheticity and different production structures, build the intuition behind these results and anchor the quantification. All of these objects are computable from three measurable ingredients: an input–output table, preference shares, and committed expenditure shares.

The second part of the paper quantifies these theoretical insights across countries. I construct the three ingredients for the 80 economies in the OECD Input–Output tables, which cover a wide range of poor and rich countries, from Pakistan, Nigeria, and India to the US, with non-homotheticity terms anchored to a six-decade US expenditure series, following [Herrendorf, Rogerson, and Valentinyi \(2013\)](#). The same estimation delivers the common preference shares. The calibrated configuration places the non-homotheticity in agriculture alone, the single-necessity case adopted by several studies in the structural change literature (e.g., [Uy, Yi, & Zhang, 2013](#); [Tombe, 2015](#); [Świecki, 2017](#)). Preferences are held common across countries so that the comparison stays transparent: poor economies differ only because the same committed quantity absorbs more of a smaller budget at their prices.⁴

In the calibrated model, I then conduct the following counterfactual exercise: the same 10 percent productivity decline in each sector, one sector at a time, in every economy, with a particular focus on the agricultural shock. The quantification yields three main results. The first is a measurement result. Against welfare, the observed Domar weight understates the response by a factor of 5.41 at the poorest-quartile median. Against GDP, it overstates the response: for almost all included economies, the true GDP response is below the observed-Domar-weight prediction. Both discrepancies are largest in the poorest countries. The second is a develop-

³Non-homothetic preferences are central to understanding economic development, in the cross-section and over time (see [Herrendorf, Rogerson, & Valentinyi, 2014](#), for a survey): consumption expenditure shares in agriculture are high in poor countries and low in rich ones, and they fall as GDP per capita grows. Engel’s law, the income effect behind these patterns, is what the non-homothetic final-demand aggregator adds to the model of production networks.

⁴For 6 of the 80 economies, the anchored committed bundle exceeds household spending outright. These economies are excluded from the baseline experiments, leaving an included sample of 74 economies.

ment result, the answer to the paper’s motivating comparison. The same agricultural shock produces GDP losses only modestly larger in poor countries, with quartile medians a factor of 1.46 apart, but welfare losses 19.0 times larger across the same divide. At the country level, the welfare cost of the shock is 2.3 times its GDP cost in the US, 6.7 times in Mexico, and 30.2 times in India.

The third is a mechanism result. Where do these differences come from: subsistence demand or the production network? The closed-form solution permits an answer through counterfactuals that switch each mechanism off separately in every economy, and through a decomposition of every response into demand, network, and interaction terms (Section 5.3). The demand side determines which countries are vulnerable to sectoral shocks: subsistence committedness by itself accounts for most of the cross-country differences, while the network by itself plays a more limited role. That does not make the network irrelevant. At every income level, it carries roughly a constant third of the welfare response and multiplies every economy’s losses by a similar factor. What changes along the development path is not whether the network matters but what it does. In poor countries, the network matters almost entirely by amplifying subsistence needs: its contribution runs through the interaction with committed consumption, which multiplies every agricultural shock. In rich countries, it matters as cost transmission: the ordinary supply-chain channel by which sectoral shocks reach the discretionary consumption basket. Poor countries’ networks amplify needs, while rich countries’ networks transmit costs. This rotation has a measured counterpart in the input–output tables themselves. Along the development path, agriculture’s purchased intermediate-input basket modernizes, with manufactured and service inputs replacing value added, the role of intermediate inputs highlighted in the literature on cross-country agricultural productivity differences (e.g., Restuccia, Yang, & Zhu, 2008; Donovan, 2021; Boppart, Kiernan, Krusell, & Malmberg, 2026).

Both ingredients are thus essential, and the paper measures their distinct roles one by one. Demand determines which countries are vulnerable, while the network multiplies the response, decides which sectors’ shocks are devastating and where, and sets how quickly repeated adverse shocks push an economy toward its subsistence threshold. A homothetic network model misses the cross-country differences entirely, while a network-free subsistence model misses the amplification and misses which shocks matter. In this precise sense, cross-country differences in the impacts of sectoral shocks are a joint product of non-homothetic demand and the production network.

Related literature. First, this paper contributes to a growing body of literature that emphasizes the role of production networks in propagating shocks and transforming micro-level shocks into macroeconomic fluctuations (e.g., Acemoglu et al., 2012; Acemoglu, Akcigit, & Kerr, 2016; Acemoglu, Ozdaglar, & Tahbaz-Salehi, 2017; Grassi, 2017; Baqaee, 2018; Baqaee & Farhi, 2019; Liu, 2019; Baqaee & Farhi, 2020; Bigio & La’O, 2020; Altinoglu, 2021; Carvalho, Nirei, Saito,

& Tahbaz-Salehi, 2021; Taschereau-Dumouchel, 2026).⁵ See Carvalho and Tahbaz-Salehi (2019) and Baqaee and Rubbo (2023) for recent surveys of this literature. Relative to this literature, this paper shifts the focus from within-country fluctuations to cross-country differences, and from GDP effects alone to the divergence between GDP and welfare impacts. I examine how production networks shape the effects of sectoral shocks on living standards when part of consumption is committed, and how this role changes as countries become richer.⁶

The closest paper is Baqaee and Burstein (2023), who study welfare measurement in the presence of income effects and taste shocks, and relate money-metric welfare to chain-weighted real GDP.⁷ This paper’s measurement layer takes up the same question for the cross-country comparison. Fixed-basket GDP is exactly log-linear in our model and does not reflect non-homothetic demand, whereas chained GDP satisfies Hulten’s theorem locally but responds nonlinearly to finite shocks and is path-dependent. Welfare satisfies Hulten’s theorem, normalized by discretionary expenditure. Whether Hulten’s theorem holds thus depends on the measurement of GDP and welfare. Unlike Baqaee and Burstein (2023), however, I analyze why the same sectoral shock has vastly different consequences in poor and rich countries, and show that the median welfare loss is 19.0 times larger in the poorest income quartile than in the richest, against a factor of 1.46 for GDP.

Second, this paper relates to the large literature that studies the sources of structural change, particularly work that emphasizes the role of non-homothetic demand in growth and development (e.g., Echevarria, 1997; Laitner, 2000; Caselli & Coleman, 2001; Kongsamut, Rebelo, & Xie, 2001; Duarte & Restuccia, 2010; Boppart, 2014; Matsuyama, 2019; Comin, Lashkari, & Mestieri, 2021; Oberfield, 2023); see Matsuyama (2023) for a survey of non-homothetic preferences.^{8,9} These papers emphasize Engel’s law as the driver of the reallocation of economic activity that

⁵A related quantitative literature examines the importance of idiosyncratic industry-level shocks propagating through networks to generate industrial comovement (e.g., Foerster, Sarte, & Watson, 2011; Di Giovanni, Levchenko, & Mejean, 2014; Atalay, 2017). See Horvath (1998, 2000) and Dupor (1999) for earlier attempts.

⁶A complementary body of work studies the role of input–output linkages in growth and development (e.g., Ciccone, 2002; Jones, 2011, 2013; Bartelme & Gorodnichenko, 2015; Oberfield, 2018; Acemoglu & Azar, 2020; Boehm & Oberfield, 2020).

⁷Relatedly, Baqaee and Farhi (2018) characterize how sectoral shocks propagate through efficient input–output economies with general, possibly non-homothetic demand, deriving first-order sufficient statistics. This paper solves its special case exactly and globally and uses it to study cross-country differences, which are not their focus. On the measurement side, Jaravel and Lashkari (2024) develop a nonparametric method for measuring welfare changes under non-homothetic preferences without estimating substitution elasticities, assuming preference heterogeneity is observable.

⁸Other papers highlight different forces as drivers of structural change: Baumol (1967), Ngai and Pissarides (2007), and Acemoglu and Guerrieri (2008) on differential productivity growth across sectors; Matsuyama (2009) on trade; and Buera and Kaboski (2009) and Herrendorf et al. (2013) for evaluations of the leading explanations. See Herrendorf et al. (2014) for a survey.

⁹A related strand of the literature studies agriculture’s role in economic development and structural change (e.g., Matsuyama, 1992; Gollin, Parente, & Rogerson, 2002, 2007; Alvarez-Cuadrado & Poschke, 2011; Bustos, Caprettini, & Ponticelli, 2016).

accompanies income growth. Relative to this literature, the paper shifts the role of Engel’s law from explaining sectoral reallocation to amplifying the effects of sectoral shocks. The structure of input–output linkages governs how shocks propagate and determines which shocks are most strongly amplified by committed consumption.

Finally, this paper relates to the literature that explores the interaction between volatility and development (e.g., [Imbs & Wacziarg, 2003](#); [Koren & Tenreyro, 2007](#); [Moro, 2012](#); [Carvalho & Gabaix, 2013](#); [Koren & Tenreyro, 2013](#); [Moro, 2015](#)). These studies highlight the non-monotonic relationship between aggregate volatility and economic development, driven by how economies diversify and specialize in certain industries along the development path. This paper complements these studies by showing that the sources of vulnerability to sectoral shocks themselves change along the development path. The roles of both non-homotheticity and production networks transform with development, and the network’s job rotates from amplifying subsistence needs in poor countries to transmitting input costs in rich ones.

Roadmap. Section 2 presents the model, its measurement conventions, and the closed-form solution. Section 3 contains the theoretical results: the responses of prices, expenditure shares, and Domar weights; the GDP and welfare theorems; the example economies; and the relation to alternative measures of GDP and welfare. Section 4 describes the data and calibration. Section 5 presents the quantitative results, and Section 6 concludes. The proofs and some additional results are provided in the Appendix.

2 Baseline Model

This section presents the baseline model, which embeds a non-homothetic final-demand system within a multisector general equilibrium framework that features an input–output network. The environment closely follows the model of [Acemoglu et al. \(2012\)](#), with a single departure: the final-demand aggregator is non-homothetic. This single departure delivers all of the paper’s results. Section 2.1 describes the environment, Section 2.2 defines the equilibrium and explains the measurement conventions in this paper, and Section 2.3 solves the equilibrium in closed form.

2.1 Environment

I consider a static economy populated by a representative household and n competitive industries, indexed by $i \in \{1, \dots, n\}$, each producing a distinct good. Each good can be consumed by the household or used as an intermediate input by other industries.

Preferences. The representative household has preferences over final consumption,

$$U = \mathcal{C}(c_1, \dots, c_n) = \prod_{i=1}^n \kappa_i (c_i - \bar{c}_i)^{\beta_i}, \quad \sum_{i=1}^n \beta_i = 1, \quad (1)$$

where c_i is consumption of good i , $\beta_i > 0$ is the preference share of good i , and $\kappa_i := \beta_i^{-\beta_i}$ is a normalization constant with no bearing on the results. The parameters $\bar{c}_i \geq 0$ are the non-homotheticity terms: good i is a necessity when $\bar{c}_i > 0$, in which case $\bar{c}_i$ is a subsistence requirement that consumption must exceed. It is a luxury when $\bar{c}_i < 0$, in which case $-\bar{c}_i$ can be interpreted as home production of the good. Demand for good i is homothetic when $\bar{c}_i = 0$.¹⁰

Budget constraint. The household is endowed with one unit of labor and supplies it inelastically. Labor is freely mobile across industries, so a single market wage w prevails. The budget constraint is given by:

$$\sum_{i=1}^n p_i c_i = w, \quad (2)$$

where p_i is the price of good i .

Technology. Each industry employs a Cobb–Douglas technology with constant returns to scale:

$$y_i = A_i \zeta_i h_i^{\alpha_i} \prod_{j=1}^n x_{ij}^{a_{ij}}, \quad (3)$$

where y_i is output of industry i , A_i is Hicks-neutral productivity, h_i is labor, x_{ij} is the quantity of good j used by industry i , α_i is the labor share, a_{ij} is the input share of good j in industry i 's production, and $\zeta_i = \alpha_i^{-\alpha_i} \prod_j a_{ij}^{-a_{ij}}$ is a normalization constant with no bearing on the results. Constant returns to scale require $\alpha_i + \sum_j a_{ij} = 1$ for all i .

The exponents $(a_{ij})_{j=1}^n$ describe industry i 's production recipe: a larger a_{ij} means that good j is a more important input into the production of good i , and $a_{ij} = 0$ means that industry i cannot usefully employ good j at any prices. Because technologies are Cobb–Douglas and markets are competitive, a_{ij} equals industry i 's expenditure on input j as a share of its sales, so the matrix $\Omega = [a_{ij}]$ corresponds directly to the entries of an input–output table.

The representative firm in each industry maximizes profits,

$$p_i y_i - w h_i - \sum_{j=1}^n p_j x_{ij}, \quad (4)$$

taking prices as given.

Throughout, I maintain the following regularity conditions.

Assumption 1. (i) For every industry i : $\alpha_i \in (0, 1)$, $a_{ij} \geq 0$ for all j , and $\alpha_i + \sum_j a_{ij} = 1$. (ii) For every good i : $\beta_i > 0$, with $\sum_i \beta_i = 1$ and $n \geq 2$.

Under Assumption 1 (i), the row sums of Ω equal $1 - \alpha_i < 1$, so the Leontief inverse $L := (I - \Omega)^{-1}$ is well defined and nonnegative. Its entries play a central role below: while a_{ij}

¹⁰The Stone–Geary preferences are the workhorse non-homothetic demand system of the structural change literature (e.g., Caselli & Coleman, 2001; Kongsamut et al., 2001; Herrendorf et al., 2013). When $\bar{c}_i = 0$ for all i , (1) reduces to the Cobb–Douglas aggregator of Acemoglu et al. (2012), and the model nests theirs.

records the *direct* exposure of industry i to good j , the entry ℓ_{ij} of L records industry i 's *direct and indirect* exposure to industry j through the network.¹¹

2.2 Equilibrium

Definition 1. A competitive equilibrium consists of allocations $(c_i, y_i, h_i, (x_{ij})_{j=1}^n)_{i=1}^n$ and prices $((p_i)_{i=1}^n, w)$ such that (i) the household maximizes (1) subject to (2), taking prices as given; (ii) each industry maximizes profits (4) subject to (3), taking prices as given; and (iii) all markets clear:

$$c_i + \sum_{j=1}^n x_{ji} = y_i \text{ for all } i \quad \text{and} \quad \sum_{i=1}^n h_i = 1. \quad (5)$$

Measurement. Two conventions fix what “GDP” and “welfare” mean in this paper. First, I measure GDP by the income approach: because labor is the only primary factor, GDP equals the household’s labor income, $\text{GDP} = w$. Second, I choose the consumption basket as the numéraire, normalizing the price index of the aggregator to one: $\prod_{i=1}^n p_i^{\beta_i} = 1$. Welfare is measured by the consumption aggregate $\mathcal{C}$ itself. These are the two natural aggregates of this economy: the income it generates and the living standard its household attains. The analysis in Section 3 characterizes both.¹²

The equilibrium is efficient: markets are competitive, so the first welfare theorem applies and the equilibrium allocation maximizes $\mathcal{C}$ on the economy’s production possibility frontier. This observation underlies the welfare results of Section 3.

2.3 The Equilibrium in Closed Form

The model admits an exact closed-form solution, which this section characterizes. To this end, I introduce three objects. First, let $q_i := p_i/w$ denote the price of good i relative to the wage. Second, define

$$u_i := \frac{p_i \bar{c}_i}{w} = \bar{c}_i q_i \quad \text{and} \quad \sigma := 1 - \sum_{k=1}^n u_k. \quad (6)$$

The term u_i is the fraction of income that the household’s requirement for good i absorbs at current prices, and σ is the fraction of income left over after all committed spending; that is, the household’s discretionary budget as a share of income. In what follows, I call u_i and σ the *committed expenditure share* of good i and the *discretionary share*, respectively. Third, define

$$\lambda^h := L^\top \beta, \quad \text{or} \quad \lambda_j^h := \sum_{i=1}^n \beta_i \ell_{ij}.$$

¹¹Expanding $L = \sum_{k=0}^\infty \Omega^k$ gives $\ell_{ij} = \mathbf{1}_{\{i=j\}} + a_{ij} + \sum_k a_{ik} a_{kj} + \dots$: the second term captures j ’s role as a direct supplier to i , the third its role as a supplier to i ’s suppliers, and so on.

¹²Section 3.4 discusses these conventions in detail and relates the paper’s two measures to alternative concepts of GDP and welfare, including chain-weighted (Divisia) real GDP and money-metric welfare.

The term λ_j^h is the sales share of industry j that would prevail in the *homothetic benchmark* of the economy; that is, the same economy with all non-homotheticity terms set to zero ($\bar{c}_i = 0$ for all i).

I maintain the following admissibility condition, which guarantees that the household can afford its committed bundle, with room to spare, at the productivity vector under consideration.

Assumption 2. *Productivities are sufficiently high that the committed bundle costs less than income, $\sum_k \bar{c}_k \prod_j A_j^{-\ell_{kj}} < 1$, and that, for every good i with $\bar{c}_i < 0$, $\beta_i(1 - \sum_k \bar{c}_k \prod_j A_j^{-\ell_{kj}}) > -\bar{c}_i \prod_j A_j^{-\ell_{ij}}$.*

In equilibrium, the two conditions are $\sigma > 0$ and interior consumption ($c_i > 0$ for every i): Proposition 1 shows that relative prices equal $\prod_j A_j^{-\ell_{ij}}$, so Assumption 2 restricts primitives only. When all non-homotheticity terms are nonnegative ($\bar{c}_i \geq 0$ for all i), the interiority condition $c_i > 0$ is implied by $\sigma > 0$, so Assumption 2 reduces to the requirement that committed spending not exhaust income.¹³

Proposition 1 (Equilibrium in closed form). *Suppose Assumptions 1 and 2 hold. Then a competitive equilibrium exists, is unique, and is characterized as follows.*

(i) **Relative prices.** *Prices relative to the wage depend on the supply side alone:*

$$\log q_i = - \sum_{j=1}^n \ell_{ij} \log A_j. \quad (7)$$

(ii) **Expenditure shares.** *The household's expenditure share on good i is:*

$$\tilde{\beta}_i := \frac{p_i c_i}{w} = \sigma \beta_i + u_i. \quad (8)$$

(iii) **Domar weights.** *The Domar weight of industry i , defined as $\lambda_i := p_i y_i / \text{GDP}$, is:*

$$\lambda = \mathbf{L}^\top \tilde{\beta} = \sigma \lambda^h + \mathbf{L}^\top \mathbf{u}, \quad \text{and thus,} \quad \lambda_j - \lambda_j^h = \sum_{i=1}^n u_i (\ell_{ij} - \lambda_j^h). \quad (9)$$

(iv) **GDP.** *Equilibrium GDP and prices satisfy*

$$\log \text{GDP} = \sum_{j=1}^n \lambda_j^h \log A_j, \quad \log p_i = \sum_{j=1}^n (\lambda_j^h - \ell_{ij}) \log A_j. \quad (10)$$

(v) **Welfare.** *Equilibrium welfare equals the discretionary budget in consumption units:*

$$\mathcal{C} = \sigma \cdot \text{GDP}, \quad \text{or} \quad \log \mathcal{C} = \sum_{j=1}^n \lambda_j^h \log A_j + \log \sigma. \quad (11)$$

¹³Interiority can bind separately only for luxuries ($\bar{c}_i < 0$), whose market consumption $c_i = \bar{c}_i + \beta_i \sigma w / p_i$ can reach zero when the discretionary budget is small. Assumption 2 is not innocuous near subsistence, and the quantitative analysis takes it seriously: Section 4 reports the economies for which it fails at the calibrated parameters.

Proof. See Appendix A.2. □

There are four remarks on Proposition 1. First, the model nests Acemoglu et al. (2012). With $\bar{c}_i = 0$ for all i , committed shares vanish ($\sigma = 1$), expenditure shares equal the constant preference shares ($\tilde{\beta}_i = \beta_i$), Domar weights collapse to the constant Domar weights under the homothetic benchmark ($\lambda = \lambda^h$), and the welfare measure coincides with GDP. Every difference between this economy and the homothetic benchmark traces back to the committed shares $\mathbf{u}$.

Second, expenditure shares, and hence Domar weights, are equilibrium objects that move with productivity. Equation (8) decomposes the household's spending on good i into a committed tier u_i , which depends on prices but not on the preference shares, and a discretionary tier $\sigma\beta_i$, which allocates the remaining budget in the fixed proportions β . In the language of the linear-expenditure-system literature, β_i is good i 's *marginal budget share*: the share of each additional unit of total spending that goes to good i . Because a productivity change moves relative prices, it revalues the committed bundle, reallocates the budget between the two tiers, and thereby shifts expenditure and sales shares throughout the network. The model generates Engel-style structural change in response to productivity growth, and Section 3.1 characterizes it.

Third, equation (9) separates the observed Domar weight into the Domar weight under the homothetic benchmark and a wedge determined entirely by the committed shares. The wedge for industry j , $\sum_i u_i(\ell_{ij} - \lambda_j^h)$, is positive when industry j is more important as a supplier to the committed bundle than to the discretionary consumption basket. In a poor economy where food, the necessity, carries a large committed position, the food industry's sales overstate the Domar weight under the homothetic benchmark by this wedge, an observation at the heart of Section 3.

Fourth, the solution reveals the model's empirical sufficient statistic. Every object in equations (6)–(11) is a function of three ingredients, and so, as Section 3 shows, is every first- and second-order response of GDP and welfare: the input–output matrix (through L), the preference shares β , and the committed expenditure shares $\mathbf{u}$. Section 4 builds these three ingredients for each economy in the sample.

3 The Impacts of Sectoral Shocks

This section characterizes the impact of sectoral productivity shocks on prices, expenditure shares, and Domar weights (Section 3.1), and on GDP and welfare (Section 3.2), the paper's main theoretical results. Section 3.3 develops the intuition in a sequence of example economies, centered on the single-necessity configuration that the quantitative analysis takes to the data. Section 3.4 relates the paper's two measures to alternative concepts of real GDP and welfare.

Because the equilibrium of Proposition 1 is an explicit smooth function of $\log A$, all differential statements below are derivatives of exact solutions. They hold at every admissible

productivity vector, not only in the neighborhood of a particular point. Throughout, a shock is a vector $d \log A := (d \log A_1, \dots, d \log A_n)^\top$, and the response to a shock to sector j alone is the coefficient on $d \log A_j$. All proofs are in Appendix A.

3.1 Building Blocks: Prices, Expenditure Shares, and Domar Weights

For a shock to sector j , define

$$m_j := \sum_{k=1}^n u_k \ell_{kj},$$

which sums, across committed positions k , the income share committed to good k times the (k, j) element of the Leontief inverse, good k 's total price exposure to sector j (equation (7)). The term m_j measures how strongly productivity in sector j revalues the committed bundle and thereby releases discretionary budget: it is large when sector j sits upstream of goods on which large income shares are committed.¹⁴

I first characterize the impact of sectoral productivity shocks on relative prices, the discretionary share, and expenditure shares.

Proposition 2 (Responses of prices, the discretionary share, and expenditure shares). *At every admissible A ,*

$$d \log q_k = - \sum_{j=1}^n \ell_{kj} d \log A_j, \quad (12)$$

$$d \sigma = \sum_{j=1}^n m_j d \log A_j, \quad (13)$$

$$d \tilde{\beta}_i = \sum_{j=1}^n (\beta_i m_j - u_i \ell_{ij}) d \log A_j, \quad (14)$$

and the share responses satisfy the adding-up restriction $\sum_i d \tilde{\beta}_i = 0$.

Proof. See Appendix A.3. □

The three formulas describe one causal chain. A positive shock to sector j lowers the relative price of every good k by k 's total reliance on j as a supplier, ℓ_{kj} . Cheaper goods shrink the income share that the committed bundle absorbs, releasing m_j units of discretionary budget per unit shock, which is an income effect. The released budget is then spent in the fixed proportions β , so each expenditure share gains a discretionary term $\beta_i m_j$ and loses the revaluation of its own committed position, $u_i \ell_{ij}$. Importantly, with Cobb–Douglas discretionary spending,

¹⁴The interpretation of m_j as measuring the release of discretionary budget is exact under pure subsistence ($\bar{c} \geq 0$, and hence $u \geq 0$), in which case $m_j \geq 0$. In the general model, negative committed positions make m_j a signed revaluation. A favorable shock to a sector upstream of home-produced goods devalues the household's implicit position there and can, on net, commit rather than release budget.

relative-price changes leave discretionary shares unchanged, so every movement in the composition of demand is an income effect operating through the committed bundle. This is why all comparative statics in this paper carry the committed shares $\mathbf{u}$, and vanish in the homothetic benchmark.

The next result translates the demand shifts into industry sales.

Theorem 1 (The impact of sectoral shocks on Domar weights). *At every admissible $\mathbf{A}$,*

$$d\lambda_i = \sum_{j=1}^n \left[\underbrace{\lambda_i^h m_j}_{\text{discretionary expansion, spread by } \beta} - \underbrace{\sum_{k=1}^n u_k \ell_{kj} \ell_{ki}}_{\text{contraction of committed demand exposed to } j} \right] d\log A_j = \sum_{j=1}^n \sum_{k=1}^n u_k \ell_{kj} (\lambda_i^h - \ell_{ki}) d\log A_j. \quad (15)$$

Moreover: (i) $\sum_i \alpha_i d\lambda_i = 0$: shocks reallocate value-added-weighted sales across industries without changing their total; and (ii) if $\bar{\mathbf{c}} = \mathbf{0}$, then $\lambda = \lambda^h$ is constant and $d\lambda = \mathbf{0}$.

Proof. See Appendix A.4. □

Theorem 1 describes the reallocation of sales shares driven by Engel’s law operating through the production network. Since Cobb–Douglas technologies fix all input–output coefficients, sales can change only because the composition of final demand changes. The response of industry i ’s sales has two parts: (i) it gains from the economy-wide discretionary expansion: the shock releases income m_j from the committed bundle, the household spends the release in the fixed proportions β , and the extra spending reaches industry i , directly and through supply chains, in proportion to its homothetic-benchmark weight λ_i^h ; and (ii) it loses the upstream sales that the shrinking committed purchases used to generate. The net effect against committed good k is proportional to $\lambda_i^h - \ell_{ki}$: industry i ’s importance to the discretionary consumption basket versus its importance to good k ’s supply chain. An industry that mainly serves subsistence-good supply chains shrinks as the economy grows richer and expands as the economy is impoverished.

There are three remarks on this result. First, the movement of Domar weights is a first-order effect of the shock: when $\bar{\mathbf{c}} \neq \mathbf{0}$, the response matrix in (15) is nonzero at every admissible state (Appendix A). This contrasts with the homothetic benchmark, where Domar weights do not respond to productivity at all (Example 1). Second, none of this movement survives into the first-order behavior of the economy’s aggregates: Section 3.2 shows that the GDP response is governed by constant coefficients, and the welfare response by the level of the Domar weights, not their changes. Those changes matter for aggregates only at second order (Appendix C). In the homothetic economies of Baqaee and Farhi (2019), these two facts could not coexist: there, the second-order impact of a shock on GDP is the response of its Domar weight, so Domar weight movements and aggregate nonlinearity are two sides of the same coin. Here, the two decouple: fixed-weight GDP is globally log-linear while the Domar weights move at first order,

and the movement resurfaces instead in the curvature of the chained index and of welfare (Section 3.4; Appendix C). Third, the share-propagation formula (15) is not entirely new, and it is the Stone–Geary case of the non-homothetic propagation equation derived by Baqaee and Farhi (2018). What the specialization to this paper’s model adds is twofold: (i) an exact solution in levels, since equation (9) holds at every productivity vector and not only as a differential; and (ii) unconditional sign results, which rest on the following property of the Leontief inverse.

Lemma 1 (Column diagonal dominance). *Under Assumption 1 (i), $\ell_{ij} \leq (1 - \alpha_i) \ell_{jj}$ for every $i \neq j$. Consequently, for every j with $\beta_j < 1$,*

$$\lambda_j^h < \ell_{jj}.$$

Proof. See Appendix A.5. □

No industry is more important to any other industry’s supply chain than it is to its own. Every unit of exposure of industry i to industry j must pass through i ’s intermediate-input bill, which caps it at $(1 - \alpha_i) \ell_{jj}$. Averaging over the discretionary consumption basket then places λ_j^h strictly below the own-multiplier ℓ_{jj} . The lemma is what makes sign results possible. In general, the direction of a Domar-weight response depends on the details of the network: by (15), industry i ’s weight rises or falls with a shock according to the sign of $\lambda_i^h - \ell_{ki}$, that is, according to whether industry i matters more to the discretionary basket or to committed good k ’s supply chain, and that comparison can go either way. For the necessity’s own weight, it cannot: the lemma places λ_j^h below ℓ_{jj} for every admissible input–output matrix, so in Corollary 1 and Section 3.3 the response of the necessity’s own Domar weight is signed unconditionally, with no restriction on the network required. The same inequality signs the luxury case, with the direction reversed by the sign of the committed position (Corollary 1; Example 3).

3.2 Main Results: The Responses of GDP and Welfare

I now turn to the aggregates. The two main results characterize the first-order responses of GDP and welfare to arbitrary sectoral shocks, and they could hardly be more different.

Theorem 2 (The impact of sectoral shocks on GDP). *At every admissible A ,*

$$d \log GDP = \sum_{j=1}^n \lambda_j^h d \log A_j,$$

a constant vector of elasticities. The formula is globally exact: for any two admissible productivity vectors A and A^0 , $\log GDP(A) - \log GDP(A^0) = \sum_j \lambda_j^h (\log A_j - \log A_j^0)$.

Proof. Immediate from (10), which is linear in $\log A$ with coefficients λ^h . □

GDP responds to sectoral shocks with the Domar weights of the *homothetic benchmark*, λ^h , not with its own observed Domar weights, λ . The next corollary draws the consequence.

Corollary 1 (Hulten’s theorem fails with respect to GDP). *Read as “the observed Domar weight is a sufficient statistic for the first-order GDP impact of a sectoral shock,” Hulten’s theorem fails generically when $\bar{\mathbf{c}} \neq \mathbf{0}$: the gap between the observed Domar weight and the GDP response is the wedge of equation (9),*

$$\lambda_j - \frac{d \log \text{GDP}}{d \log A_j} = \lambda_j - \lambda_j^h = \sum_{i=1}^n u_i (\ell_{ij} - \lambda_j^h), \quad (16)$$

which vanishes for all j simultaneously only in the knife-edge case in which committed value shares are proportional to the preference shares, $\mathbf{u} = (\sum_k u_k) \boldsymbol{\beta}$. In particular, if good F is a necessity ($\bar{c}_F > 0$, $\bar{c}_k = 0$ otherwise) and $\beta_F < 1$, then by Lemma 1,

$$\lambda_F - \lambda_F^h = u_F (\ell_{FF} - \lambda_F^h) > 0: \quad (17)$$

the observed Domar weight of the necessity strictly overstates the GDP impact of its productivity shock, unconditionally. In contrast, if good V is a luxury ($\bar{c}_V < 0$, $\bar{c}_k = 0$ otherwise) and $\beta_V < 1$, the same inequality flips the wedge: $\lambda_V - \lambda_V^h = u_V (\ell_{VV} - \lambda_V^h) < 0$, so the observed Domar weight understates the GDP impact of the luxury’s productivity shock.

Proof. See Appendix A.6. □

Two readings explain the pair of results. The first is technological. Under the income approach with a fixed-weight numéraire, $\log \text{GDP} = -\sum_i \beta_i \log q_i$ is a fixed-direction measure of the economy’s production possibilities, and relative prices $\mathbf{q}$ depend on the supply side alone (see equation (7)). A shock to sector j pushes the frontier outward good-by-good at rates ℓ_{ij} , and averaging those rates with the fixed weights β_i gives λ_j^h . The position of demand on the frontier, which is all that non-homotheticity determines, never enters the calculation. In this precise sense, non-homotheticity drops out of the first-order GDP calculation: the income effects that move every share in the economy (Proposition 2 and Theorem 1) leave no first-order trace in measured GDP.¹⁵

The second reading is statistical. Hulten’s sufficiency is the statement that *average* expenditure shares equal *marginal* ones. The observed Domar weight $\lambda_j = (\mathbf{L}^\top \tilde{\boldsymbol{\beta}})_j$ is built from the household’s average expenditure shares $\tilde{\boldsymbol{\beta}}$, while the GDP response $\lambda_j^h = (\mathbf{L}^\top \boldsymbol{\beta})_j$ is built from the marginal shares $\boldsymbol{\beta}$. In the homothetic case, the two coincide, which is Hulten’s theorem. However, with committed consumption, the average tilts toward the committed bundle, and sales data alone can no longer reveal the GDP response: recovering λ_j^h requires the network $\mathbf{L}$ and the marginal shares, an estimated object. Two economies with identical input–output tables and identical observed Domar weights, but different committed positions, respond differently to the same shock. For the necessity itself, the direction is unambiguous: committed spending

¹⁵They leave no trace at any order. $\log \text{GDP}$ is globally linear in $\log \mathbf{A}$, so this measure of GDP exhibits no amplification, no asymmetry between positive and negative shocks, and no sensitivity to volatility in log units, for shocks of any size. Appendix C develops the second-order layer of the model, where the interesting curvature belongs to welfare.

on food, the necessity, inflates food-chain sales throughout the network, so the observed Domar weight of the food industry overstates the GDP consequences of productivity shocks to it. In the quantification of Section 5, the median ratio of the GDP response to the observed Domar weight is 0.20 in the poorest income quartile of the quantitative sample.

The welfare response takes a very different form.

Theorem 3 (The impact of sectoral shocks on welfare). *At every admissible $\mathbf{A}$,*

$$d \log \mathcal{C} = \sum_{j=1}^n \frac{\lambda_j}{\sigma} d \log A_j, \quad \text{with} \quad \frac{\lambda_j}{\sigma} = \frac{p_j y_j}{GDP - \sum_k p_k \bar{c}_k}. \quad (18)$$

Hulten's theorem holds for welfare, with the Domar weight normalized by the discretionary share. Equivalently,

$$d \log \mathcal{C} = \underbrace{d \log GDP}_{=\sum_j \lambda_j^h d \log A_j} + \underbrace{\frac{d \sigma}{\sigma}}_{=\sum_j (m_j / \sigma) d \log A_j}. \quad (19)$$

The welfare impact is the GDP impact plus a subsistence-relief term that GDP does not record.

Proof. See Appendix A.7. □

Corollary 2 (Sufficiency up to a scalar). *The observed Domar weights are sufficient statistics for relative welfare impacts across shocked sectors, and for absolute impacts up to the single country-specific scalar σ . Whenever net committed expenditure is positive ($\sigma < 1$), every sectoral shock has a welfare impact strictly larger than its observed Domar weight, scaled by $1/\sigma > 1$. Relative to the GDP impact,*

$$\frac{d \log \mathcal{C}}{d \log A_j} - \frac{d \log GDP}{d \log A_j} = \frac{m_j}{\sigma},$$

which, in an economy with committed positions $\mathbf{u} \geq \mathbf{0}$, is strictly positive for every sector j that supplies some necessity's chain, directly or indirectly.

Proof. Immediate from Theorem 3 and equation (13). □

Why does Hulten's logic survive for welfare when it fails for GDP? Hulten's argument has three ingredients: efficiency, the envelope theorem, and an Euler step that converts marginal utility into a money denominator. The first two are intact here. The equilibrium is efficient, and the envelope theorem makes the elaborate reallocations of Theorem 1 welfare-neutral at first order: only price changes matter, weighted by the chosen consumption point. Only the Euler step bends. Because utility is homogeneous of degree one in consumption *net of subsistence*, the value of the welfare aggregate is discretionary expenditure, $GDP - \sum_k p_k \bar{c}_k = \sigma \cdot GDP$, not GDP itself. Hulten's conclusion then survives verbatim with that one substitution: the first-order welfare value of a shock to sector j is the sector's sales expressed as a share of the welfare-relevant total.

The scalar $1/\sigma$ is the wedge between welfare accounting and GDP accounting. Theorem 3 holds for any pattern of committed positions (Example 3 works out the luxury case, where the wedge damps rather than amplifies), but its economics is sharpest under subsistence. A shock that changes full income by one percent changes *discretionary* income by $1/\sigma$ percent, because committed spending does not scale with income. Decomposition equation (19) makes the same point by channels. An adverse shock to food, the necessity, lowers real income (the GDP channel, weight λ_j^h) and raises the cost of the committed bundle relative to the wage, squeezing the discretionary budget from which all utility flows (the subsistence channel, weight m_j/σ). GDP records only the first channel. The second is largest where committed positions are large and the shocked sector is deeply embedded in their supply chains, and it is divided by σ . The poorer the economy, the smaller the discretionary budget that each unit of subsistence squeeze perturbs. As $\sigma \rightarrow 0$, the multiplier grows without bound. The quantification finds values of $1/\sigma$ as high as 8.905 at the calibrated parameters. One unit conversion locates the money metric. Expressed as an equivalent variation in endowments (or as a level change relative to *total* rather than discretionary resources), the same first-order impact carries the coefficient λ_j itself, with no $1/\sigma$. The scalar expresses shocks in the units of what the household has left to lose. Section 3.4 develops this taxonomy.

In sum, *Hulten's theorem fails with respect to GDP but holds for welfare measures up to a country-specific scalar*. The scalar is country-specific because the discretionary share is an equilibrium object. Economies where the committed bundle absorbs more of the budget carry smaller σ and larger multipliers. The GDP response is governed by the homothetic-benchmark Domar weights rather than the observed weights, which are larger for the necessity sector and for the industries more exposed to its chain than to the discretionary basket. By contrast, the welfare response is governed by the observed Domar weight scaled up by $1/\sigma$. The observed Domar weight is thus a poor proxy for both aggregates at once, and errs in opposite directions. It overstates the GDP impact of shocks to necessities and understates their welfare impact, with both discrepancies largest in poor economies. This taxonomy is the organizing principle of the quantitative analysis in Section 5. All second-order properties are characterized in Appendix C: the curvature of welfare, its global concavity under subsistence, the asymmetry between losses and gains, subsistence thresholds, and the welfare cost of volatility. The complements of Section 5.4 report how much these add at the calibrated shocks.

3.3 Example Economies

This subsection develops the theory's intuition through a sequence of example economies. The single-necessity economy of Example 2 is the paper's leading configuration, the one the quantitative analysis estimates. It is flanked by the homothetic benchmark it departs from, its mirror image with a single luxury, and a two-industry economy in which every object is explicit. A three-sector economy that combines a necessity with a luxury is worked out in Appendix D.

Example 1 (The homothetic benchmark). Let $\bar{c}_i = 0$ for all i . Then $\sigma = 1$, $\tilde{\beta}_i = \beta_i$, and $m_j = 0$: Domar weights are constants of the network, $\lambda = \lambda^h = \mathbf{L}^\top \beta$ with $d\lambda = \mathbf{0}$, and GDP and welfare are the same object, obeying Hulten's formula globally:

$$\log \text{GDP} = \log \mathcal{C} = \sum_{j=1}^n \lambda_j \log A_j.$$

These are the results of [Acemoglu et al. \(2012\)](#). Domar weights are constants of the network, sales shares are sufficient statistics for the impact of any sectoral shock, and, because the formula is globally linear, Hulten's theorem holds globally, not only locally. Every deviation from this benchmark in the examples below is proportional to the committed positions u_k . The gap between observed and preference shares is the model's single new force.

The leading example is the minimal departure from Example 1: one good is a necessity, and demand for all others is homothetic. This configuration has direct precedents in the structural change literature: in three-sector models, [Uy et al. \(2013\)](#) calibrate the subsistence term in agriculture alone, [Świecki \(2017\)](#) imposes it as the baseline, and [Tombe \(2015\)](#) adopts the same demand side. The two-sector tradition places the same term in agriculture against a non-agricultural composite (e.g., [Gollin et al., 2002](#); [Restuccia et al., 2008](#); [Donovan, 2021](#)).

Example 2 (A single necessity). Let good F ("food") carry the only non-homotheticity term: $\bar{c}_F > 0$ and $\bar{c}_k = 0$ for $k \neq F$, with $\beta_F < 1$. Then $u_F = \bar{c}_F q_F$ is the committed food share, $\sigma = 1 - u_F$, and the general formulas specialize as follows.

- (i) **Expenditure shares.** For a shock to any sector j , $d\tilde{\beta}_F = -(1 - \beta_F) u_F \ell_{Fj} d\log A_j \leq 0$ and $d\tilde{\beta}_s = \beta_s u_F \ell_{Fj} d\log A_j \geq 0$ for $s \neq F$ (strict whenever $\ell_{Fj} > 0$). Positive shocks lower the food share and raise all others, and adverse shocks do the reverse, which is Engel's law for necessities.
- (ii) **Domar weights.** $\lambda_j = \lambda_j^h + u_F (\ell_{Fj} - \lambda_j^h)$ for every j . Observed Domar weights exceed their homothetic-benchmark counterparts for the industries that supply the food chain, that is, for those with $\ell_{Fj} > \lambda_j^h$. For the food industry itself, the condition holds unconditionally, by (17).
- (iii) **Changes in Domar weights.** $d\lambda_i = \sum_j u_F \ell_{Fj} (\lambda_i^h - \ell_{Fi}) d\log A_j$. In particular, the food industry's own weight falls with favorable shocks and rises with adverse ones:

$$d\lambda_F = u_F \ell_{FF} (\lambda_F^h - \ell_{FF}) d\log A_F < 0 \text{ for } d\log A_F > 0, \quad (20)$$

unconditionally, by Lemma 1. The mechanism runs through the shares of item (i). An adverse food shock raises q_F , and with it the committed share u_F . The food expenditure share therefore rises while every other share falls. Passed through the network ($\lambda = \mathbf{L}^\top \tilde{\beta}$), this tilt raises the sales of food and of the industries more exposed to the food chain than

to the discretionary basket ($\ell_{Fi} > \lambda_i^h$, as in item (ii)). Equivalently, $d\lambda_F > 0$ for $d\log A_F < 0$. As the economy is impoverished, these industries grow more important in the states where food productivity is low.

(iv) **GDP and welfare.** For a shock to any sector j , $d\log \text{GDP} = \lambda_j^h d\log A_j$, while

$$d\log \mathcal{C} = \frac{\lambda_j}{\sigma} d\log A_j = \left[\lambda_j^h + \frac{u_F}{\sigma} \ell_{Fj} \right] d\log A_j. \quad (21)$$

The welfare–GDP gap for shock j is $(u_F/\sigma) \ell_{Fj}$. This is the committed food position, scaled by $1/\sigma$, times the exposure of the food price to sector j .

Three lessons carry to the quantification. First, *who* is vulnerable and *which* shocks matter are determined by different parts of the model. The size of the welfare–GDP gap is governed by the demand side, through u_F/σ , the committed position per unit of discretionary budget, which is large precisely in poor economies. Which shocks carry the gap is governed by the supply side, through the food-chain exposures ℓ_{Fj} . Without the network ($\mathbf{L} = \mathbf{I}$), only the food shock itself would carry a committed surcharge, whereas with it, every sector that feeds the food chain does, in proportion to its column in the Leontief inverse.

Second, the own-shock welfare response admits an exact decomposition into the contributions of the model’s two mechanisms. Taking $j = F$ in (21) and splitting each term against the no-network benchmark ($\mathbf{L} = \mathbf{I}$, so that $\lambda_F^h = \beta_F$ and $\ell_{FF} = 1$),

$$d\log \mathcal{C} = \left[\underbrace{\beta_F}_{\text{neither}} + \underbrace{(\lambda_F^h - \beta_F)}_{\text{network only}} + \underbrace{\frac{u_F}{\sigma}}_{\text{non-homotheticity only}} + \underbrace{\frac{u_F}{\sigma} (\ell_{FF} - 1)}_{\text{interaction}} \right] d\log A_F. \quad (22)$$

The first term is the direct final-demand share, present with neither mechanism. The second is what the network adds when demand is homothetic, namely the food used indirectly in producing the discretionary consumption basket. The third is what the non-homotheticity adds without any network, the committed position itself. The fourth is the interaction: the network’s own-loop, $\ell_{FF} - 1$, multiplying the committed position. Each term is identified by the model’s sufficient statistic, and Section 5 measures all four for every economy, along the development path.

Third, the example exposes the curvature that the second-order analysis formalizes. An adverse food-chain history raises q_F , hence u_F , hence every welfare response. The committed share grows at the rate ℓ_{FF} per unit of adverse own shock, so the multiplier $1/\sigma$ itself deteriorates as shocks accumulate, at the rate $(u_F/\sigma) \ell_{FF}$. Welfare responses are therefore convex in adverse shocks. Losses exceed gains, and there is a finite shock size, a subsistence threshold, at which the committed bundle exhausts income. Appendix C characterizes this layer, and the quantification reports each economy’s distance to it.

A single luxury. Reversing the sign of the committed position reverses every conclusion.

Example 3 (A single luxury). Let good V carry the only non-homotheticity term, now with $\bar{c}_V < 0$ and $\bar{c}_k = 0$ for $k \neq V$, and $\beta_V < 1$. Then $u_V = \bar{c}_V q_V < 0$, the discretionary share exceeds one, $\sigma = 1 + |\bar{c}_V| q_V > 1$, and $m_j = u_V \ell_{Vj} \leq 0$, so every conclusion of Example 2 reverses sign.

- (i) **Expenditure shares.** For a shock to any sector j , $d\tilde{\beta}_V = -(1 - \beta_V) u_V \ell_{Vj} d\log A_j \geq 0$ and $d\tilde{\beta}_s = \beta_s u_V \ell_{Vj} d\log A_j \leq 0$ for $s \neq V$ (strict whenever $\ell_{Vj} > 0$). Positive shocks raise the luxury's expenditure share at the expense of all others.
- (ii) **Domar weights.** $\lambda_V - \lambda_V^h = u_V (\ell_{VV} - \lambda_V^h) < 0$. The observed Domar weight *understates* the GDP impact of the luxury's productivity.
- (iii) **Changes in Domar weights.** $d\lambda_V = u_V \ell_{Vj} (\lambda_V^h - \ell_{VV}) d\log A_j \geq 0$ unconditionally, by Lemma 1, and strictly whenever $\ell_{Vj} > 0$. Favorable shocks raise the luxury's weight, adverse shocks lower it, and the luxury and its suppliers shrink as the economy is impoverished, the mirror image of (20).
- (iv) **GDP and welfare.** For a shock to any sector j , $d\log \text{GDP} = \lambda_j^h d\log A_j$, as always, while $d\log \mathcal{C} = (\lambda_j/\sigma) d\log A_j$ with $\lambda_j/\sigma = \lambda_j^h + u_V \ell_{Vj}/\sigma$, below λ_j^h whenever $\ell_{Vj} > 0$. Moreover, $1/\sigma < 1$ damps every response. Welfare impacts are *smaller* than their sales-based readings.

Interpret $\bar{c}_V < 0$ as home production. Utility depends on $c_V + |\bar{c}_V|$, market purchases plus a home endowment, so the household is implicitly a seller of $|\bar{c}_V|$ units at price p_V and its discretionary income $w + |\bar{c}_V| p_V$ exceeds its wage. Two mirror images of the necessity mechanics follow. On the income side, a positive shock in V 's supply chain cheapens the luxury and thereby devalues the home endowment. The household is effectively poorer ($m_j < 0$), and expenditure tilts toward the luxury and away from all other goods: Engel's law for luxuries. On the welfare side, part of the consumption of good V is insulated from the market, so market shocks matter less for utility than for market income. Every response is damped by $1/\sigma < 1$, and further damped along V 's supply chain by the endowment revaluation. Along a path of adverse shocks, the mechanisms attenuate rather than compound, because σ rises as the economy gets poorer. This economy therefore exhibits mitigation where the subsistence economy exhibits amplification.

Two industries, fully explicit. At $n = 2$, every abstract object of the theory becomes a ratio of primitive shares.

Example 4 (Two industries with symmetric preference shares). Let $n = 2$, with F a necessity and R the rest: $\Omega = \begin{pmatrix} a_{FF} & a_{FR} \\ a_{RF} & a_{RR} \end{pmatrix}$, labor shares $\alpha_F = 1 - a_{FF} - a_{FR} > 0$ and $\alpha_R = 1 - a_{RF} - a_{RR} > 0$, preference shares $\beta = (\frac{1}{2}, \frac{1}{2})$, and $\bar{c}_F > 0 = \bar{c}_R$. Write $D := \det(I - \Omega) > 0$.

- (i) **Leontief inverse.** $L = \frac{1}{D} \begin{pmatrix} 1 - a_{RR} & a_{FR} \\ a_{RF} & 1 - a_{FF} \end{pmatrix}$, and column dominance holds with explicit gaps:

$$\ell_{FF} - \ell_{RF} = \frac{\alpha_R}{D} > 0, \quad \ell_{RR} - \ell_{FR} = \frac{\alpha_F}{D} > 0. \quad (23)$$

(ii) **Demand system.** $q_F = \exp\{-\ell_{FF}\log A_F - \ell_{FR}\log A_R\}$, $u_F = \bar{c}_F q_F$, $\sigma = 1 - u_F$, and

$$\tilde{\beta}_F = \frac{1 + u_F}{2}, \quad \tilde{\beta}_R = \frac{1 - u_F}{2} :$$

the expenditure-share gap between food and the rest is the committed share, and the two share responses are mirror images.

(iii) **Domar weights and impacts.** $\lambda_F = \lambda_F^h + u_F \frac{\alpha_R}{2D}$ and $\lambda_R = \lambda_R^h - u_F \frac{\alpha_F}{2D}$, and for a shock to either sector $j \in \{F, R\}$,

$$d\lambda_F = -u_F \ell_{Fj} \frac{\alpha_R}{2D} d\log A_j, \quad d\lambda_R = u_F \ell_{Fj} \frac{\alpha_F}{2D} d\log A_j,$$

so the food weight falls and the other industry's weight rises for positive shocks ($d\log A_j > 0$), and conversely for adverse ones, while

$$d\log \text{GDP} = \lambda_j^h d\log A_j, \quad d\log \mathcal{C} = \left[\lambda_j^h + \frac{u_F \ell_{Fj}}{1 - u_F} \right] d\log A_j.$$

Two lessons stand out. First, at $n = 2$, every sign is unconditional, with magnitudes governed by the labor shares. The dominance gaps (23) are the value-added leakages of the two recycling routes (the route $R \rightarrow F$ leaks R 's labor share α_R), so a positive food shock lowers, and an adverse one raises, the food Domar weight for any admissible Ω , with asymmetric preference shares changing magnitudes only. Genuine network-dependence of signs requires $n \geq 3$, where $\lambda_i^h \geq \ell_{Fi}$ is a real dichotomy for non-necessity industries. Second, the wedges scale with the committed share u_F , that is, with poverty, interacted with network primitives: the sales wedge with $\alpha_R/(2D)$, the welfare amplification with $\ell_{Fj}/(1 - u_F)$. A poor economy, carrying a large committed share after an adverse food-productivity history, has a visibly larger food Domar weight and a larger welfare response to both sectors' shocks (strictly, whenever the shocked sector has positive food-chain exposure, $\ell_{Fj} > 0$). Where a large committed share meets a more food-intermediated production structure (a larger a_{FF} , hence a larger ℓ_{FF} and a smaller D), the two sources of the wedge compound.

Section 5 takes the single-necessity configuration of Example 2 to the data.

3.4 Relation to Other Measures of GDP and Welfare

The results above are statements about two specific measures: GDP by the income approach under a consumption-basket numéraire, and the consumption aggregate $\mathcal{C}$ as a welfare measure. This subsection locates them among the alternatives, and explains why the paper commits to this pair. Proofs and details are in Appendices A.8 and B.

Fixed-weight deflators. Theorem 2 is not an artifact of the particular basket chosen as numéraire. For *any* fixed weight vector $\mathbf{b}$ ($b_i \geq 0$, $\sum_i b_i = 1$), nominal GDP deflated by the

geometric index $\prod_i p_i^{b_i}$ satisfies

$$d \log \text{GDP}_b = d \log w - \sum_{i=1}^n b_i d \log p_i = \sum_{j=1}^n \left(L^\top \mathbf{b} \right)_j d \log A_j. \quad (24)$$

GDP is exactly log-linear under every fixed-weight deflator, with constant elasticities $L^\top \mathbf{b}$, the Domar weights of a fictitious homothetic economy with expenditure shares $\mathbf{b}$. The paper's choice $\mathbf{b} = \beta$ selects the preference-shares member of this family, which is the homothetic benchmark's own basket. No *fixed* member tracks the observed Domar weights. The member matched to the current shares, $\mathbf{b} = \tilde{\beta}(A^0)$, responds with $\lambda(A^0)$ at A^0 itself, but its elasticities stay constant, while the observed Domar weights change (Theorem 1). Matching them everywhere requires re-weighting continuously, which is chaining.

Chain-weighted (Divisia) real GDP. Statistical agencies increasingly chain, re-weighting the deflator continuously by current expenditure shares. Setting $\mathbf{b} = \tilde{\beta}(A)$ in equation (24) delivers chain-weighted real GDP:

$$d \log Y^{\text{ch}} = d \log w - \sum_{i=1}^n \tilde{\beta}_i d \log p_i = \sum_{j=1}^n \lambda_j d \log A_j. \quad (25)$$

Chained real GDP obeys Hulten's theorem with the *observed* Domar weights, pointwise, as in the general aggregation results of [Hulten \(1978\)](#) and, for economies with arbitrary non-homothetic and unstable demand, Proposition 6 of [Baqee and Burstein \(2023\)](#), of which (25) is the single-factor, stable-preference special case. The pointwise property comes at a cost that the fixed-weight family does not pay. Because the weights $\lambda(A)$ move with the state (Theorem 1), the chained index is curved, and, because the Jacobian of the weight field is not symmetric, its discrete-horizon value generically depends on the *path* of shocks, not only on endpoints. Appendix B characterizes this path dependence in closed form, identifies the knife-edge states where it vanishes, and shows that, at empirically relevant shock sizes, it is quantitatively small. The chained index nonetheless answers an accounting question, not a counterfactual one.

Money-metric welfare. [Baqee and Burstein \(2023\)](#) measure welfare by the equivalent variation in *endowments*: the proportional change in factor supplies that would replicate a shock's welfare effect. In this economy, their metric EV^M satisfies

$$d \text{EV}^M = \sum_{j=1}^n \lambda_j d \log A_j = d \log Y^{\text{ch}}, \quad \text{while} \quad d \log \mathcal{C} = \frac{1}{\sigma} d \text{EV}^M. \quad (26)$$

The endowment metric coincides pointwise with chained real GDP, and differs from the paper's welfare measure only in its units. Both are money metrics of the same preferences. EV^M asks "how much more *endowment* would compensate?" while $\log \mathcal{C}$ asks "how much more *discretionary income* would compensate?" The household's indirect utility function is discretionary income deflated by the price index, $\mathcal{C} = (w - \sum_k p_k \bar{c}_k) / \prod_i p_i^{\beta_i}$, so $d \log \mathcal{C}$ is itself an equivalent

variation in discretionary-income units, and involves no cardinal choice beyond the choice of units.¹⁶

The paper prefers its pair of measures for two reasons. First, the units are matched to the question. EV^M answers a resource question, measuring a shock in endowment equivalents, while $d\log \mathcal{C}$ answers the living-standards question at the center of this paper, measuring it in above-subsistence consumption equivalents. The wedge between them is not created by the units. $1/\sigma$ is the elasticity of welfare with respect to income, which rises without bound near subsistence and which the endowment metric removes by construction. Near subsistence, a one-percent endowment loss still wipes out $1/\sigma$ percent of above-subsistence living standards, and it is the latter that the paper’s question is about. The second reason is practicality. Like fixed-weight GDP, $\mathcal{C}$ is path-independent and available in closed form at any shock size, whereas the chained index is not. For the cross-country counterfactuals this paper runs, the path-independent pair is the operational one.

4 Calibration

This section takes the model to the data. The empirical strategy follows directly from the sufficient-statistic property of Proposition 1. Every object the theory characterizes requires three ingredients for each economy, namely the Leontief inverse L , the preference shares β , and the committed expenditure shares $\mathbf{u}$. The quantification is the construction of these three ingredients, economy by economy, followed by the experiments of Section 5. I calibrate the single-necessity configuration of Example 2, mapping the necessity to agriculture in each economy. Section 4.1 describes the data. Section 4.2 describes the calibration and estimation. Details and sources are described in Appendix E.

¹⁶Valentinyi (2021) adopts the same normalization in a Stone–Geary network economy, noting that the Stone–Geary aggregator admits a price index that does not depend on the level of output. The choice has two justifications beyond precedent. First, a numéraire must be a function of prices alone, and $\prod_i p_i^{\beta_i}$, the marginal price of a unit of $\mathcal{C}$, is the one index this demand system offers with that property. Second, deflating both aggregates by the same index makes $\mathcal{C} = \sigma \cdot \text{GDP}$ exact, so the welfare–GDP gap vanishes in the homothetic benchmark and, away from it, reflects the committed positions alone. Any other deflator would open a gap even without subsistence, a pure artifact of mismatched units. Comin et al. (2021) and Matsuyama (2023) instead define the price index as expenditure per unit of utility, with expenditure total; applied to this economy at the equilibrium budget, their index equals $\prod_i p_i^{\beta_i} / \sigma$, the average cost of a unit of $\mathcal{C}$ with committed spending included, of which $\prod_i p_i^{\beta_i}$ is the marginal part. Applied to the discretionary budget, it returns $\prod_i p_i^{\beta_i}$ itself. Their convention folds the committed burden into a country-specific price level, while this paper carries it as the separate multiplier $1/\sigma$. The two conventions are thus the same accounting, packaged differently. Nothing in the paper’s results turns on the choice. The response coefficients λ_j^h , λ_j , and λ_j/σ are invariant to the price-index convention, and under the total-expenditure convention, the welfare theorem reappears as the response formula of that index.

4.1 Data

Input-output tables. The production-network side comes from the OECD Inter-Country Input-Output (ICIO) tables, 2025 edition, for the 2017 reference year.¹⁷ For each economy, I collapse the 50-industry table to three sectors, aggregating flows before taking coefficients and using total (domestic plus imported) cost shares, following Herrendorf et al. (2013). Appendix E details the conventions, and a domestic-flows-only variant appears among the robustness checks. The three sectors are: *A*, agriculture, comprising agriculture, forestry, and fishing together with the manufacture of food, beverages, and tobacco (ISIC A01–A03 and C10–C12); *M*, manufacturing (mining, utilities, and construction; ISIC B–F excluding C10–C12); and *S*, services, comprising all remaining activities, from trade and transport to finance, government, and personal services (ISIC G–T). Integrating food processing into *A* follows Herrendorf et al. (2013), whose consumption good “food” is the processed article that households actually buy. Uy et al. (2013) adopt the same labeling, likewise merging food processing into agriculture. From each economy’s three-sector table, I read off the input-output matrix Ω , the labor shares α , and hence $L = (I - \Omega)^{-1}$, together with the observed expenditure shares $\tilde{\beta}$ and Domar weights λ .

Prices and incomes. Cross-country food price levels come from the International Comparison Program (ICP), 2017 cycle. The household-facing consumption budgets E^c are built from the same ICIO final-demand accounts as the expenditure shares and converted to per-person terms with World Bank population data (Appendix E). A single final-demand concept, comprising household and non-profit consumption, government consumption, and residents’ direct purchases abroad, serves the expenditure shares, the budgets, and the Domar weights alike. Incomes (PPP GDP per capita, used throughout for the income axis and quartiles) come from the Penn World Table 11.0 (Feenstra, Inklaar, & Timmer, 2015). Income quartiles are quartiles of 2017 GDP per capita (PPP) across all 80 table economies, twenty per quartile. Quartile statistics are computed over the included economies in each quartile (the baseline sample described below), with means and medians reported separately.

The anchor series. The one remaining ingredient, the level of committed food consumption, cannot be read off any national account, and Section 4.2 explains why. I discipline it with a long US time series: the share of food and beverages purchased for off-premises consumption in personal consumption expenditures, from the national income and product accounts (NIPA), over 1947–2010, the sample of Herrendorf et al. (2013). The anchor reproduces their restricted (unit-elasticity) specification, which they report fits essentially as well as their unrestricted one, as a single-equation estimate on the published NIPA food line.

¹⁷The tables cover 80 economies individually (I exclude the rest-of-world aggregate), spanning a wide range of GDP per capita, from Pakistan, Nigeria, and India to the US. The conclusions below are robust to using the tables’ 2021 reference year. Section 5.4 reports this and the other robustness checks.

4.2 Calibration and Estimation

What is observed and what must be estimated. Of the three ingredients, L is data. The other two are not. The observed expenditure shares $\tilde{\beta} = \sigma\beta + \mathbf{u}$ are an equilibrium mixture of preference shares and committed shares, and no national account records the split. Worse, the split is not identified from expenditure patterns at a point in time. At a single price vector, shifting committed values along the preference shares ($u_i \mapsto u_i + s\beta_i$, $\sigma \mapsto \sigma - s$) leaves demand behavior unchanged at every income level, the classical identification problem of the linear expenditure system. Identification requires price–income variation or an external anchor. I use an anchor estimated from price–income variation: a six-decade time-series estimate of the US demand system pins down the committed quantity (that is the identification step). International price data then merely convert it into each economy’s committed share, and the preference shares follow from the US expenditure shares through the model’s own share equation.

The anchored demand system. Under the single-necessity configuration ($\bar{c}_A > 0$, $\bar{c}_M = \bar{c}_S = 0$), the model’s expenditure-share equation (8) for food collapses to a two-parameter relationship between the observed food share, food prices, and total expenditure,

$$\tilde{\beta}_{A,t} = \beta_A + (1 - \beta_A) \frac{p_{A,t} \bar{c}_A}{E_t}, \quad (27)$$

where E_t is consumption expenditure per person, the demand system’s own budget concept, and per person because the committed quantity is a per-person requirement: the aggregate share equation is invariant to population only in per-capita terms.¹⁸ Equation (27) is an identity of the model, though its assignment to a NIPA expenditure category is a measurement choice (Appendix E). Equation (27) makes the identification transparent. The preference share is the asymptote of the food share as income grows, and the committed quantity governs the slope with respect to the price-to-expenditure ratio. Engel’s law traced through time is the identifying variation. I estimate (27) on the US NIPA food-share series over 1947–2010, a window across which the price-to-expenditure ratio that identifies the committed quantity falls by a factor of about four. The equation is linear in a one-to-one reparametrization (on the interior $a < 1$, where the estimate lies), so ordinary least squares delivers the nonlinear least-squares optimum (Appendix E). Rebasing food prices so that the committed quantity is denominated directly in 2017 US dollars per person per year, the anchored estimate is $\bar{c}_A^* = \$1,847.67$, with a root-mean-square share error of 0.0057 over six decades. The anchor series’ own fitted intercept, the

¹⁸In the model, GDP, aggregate consumption expenditure, and the household budget coincide, so the theory itself is silent on which national-accounting aggregate instantiates E_t . The demand system decides. The share’s numerator and denominator must be the same budget (here, personal consumption expenditures), since a GDP denominator would mix the consumption-allocation margin, which identifies $\bar{c}_A$, with the saving margin, about which the preferences say nothing. $\sigma = 1 - u$ must be the discretionary share of the consumption budget, because the welfare aggregate is $C = \sigma E$. Estimating instead on the broader consumption-plus-government denominator moves the committed value to \$1,825.06 (Appendix F).

asymptotic food share, is $\hat{\omega}_A = 0.0316$.¹⁹

The committed value is what the anchor contributes to the calibration. The preference shares are recovered from the model's share equation itself, evaluated at the US in 2017. Under the two zero restrictions, equation (8) inverts to $\beta_A = (\tilde{\beta}_A^{\text{US}} - u_A^{\text{US}}) / \sigma^{\text{US}}$, $\beta_M = \tilde{\beta}_M^{\text{US}} / \sigma^{\text{US}}$, and $\beta_S = \tilde{\beta}_S^{\text{US}} / \sigma^{\text{US}}$, where u_A^{US} evaluates the anchored committed value against US spending. The result is $\beta = (0.0250, 0.1211, 0.8540)$, summing to one by the US share identity.²⁰ Importantly, both elements of preferences, β and the committed quantity, are held common across countries. Every cross-country difference in the model is carried by prices, incomes, and the input–output structure. This is a deliberately parsimonious design. It does not estimate anything about poor countries' tastes: it rules taste differences out by construction. A poor economy has a large committed share only because the same subsistence quantity absorbs more of a smaller budget at its prices.

From the anchor to country committed shares. For each economy c , the committed share evaluates the common committed quantity at local food prices relative to local spending:

$$u_A^c = \frac{\bar{c}_A^* \times \text{PLI}_{\text{food}}^c}{E^c}, \quad \sigma^c = 1 - u_A^c, \quad (28)$$

where $\text{PLI}_{\text{food}}^c$ is the economy's ICP food price level and E^c its household-facing consumption budget per person, built from the same ICIO accounts as the expenditure shares. The implied committed shares are a model implication of the anchored calibration, not a fitted object. The median committed food share among included economies is 0.606 in the poorest income quartile and 0.056 in the richest, and the implied welfare multiplier $1/\sigma$ declines steeply with income, reaching 8.905 at its maximum. The empirical content therefore lies in moments the calibration never targets. The remainder of this subsection reports how the implied shares compare with observed expenditure data, where they agree and where they fail. The model

¹⁹The anchor estimate is uncertain in two ways: sampling uncertainty from the regression itself, and specification uncertainty from the modeling choices around it. The specification side is the larger of the two. *Sampling*: the fit's residuals are strongly persistent (first-order autocorrelation 0.94), so standard errors are serial-correlation robust: Newey–West standard errors on $\bar{c}_A^*$ run \$39–\$53 across lag windows, and a residual block-bootstrap 95 percent interval is [\$1,763, \$1,936], about ± 2 –3 percent. *Specification*: moving the estimation end-year across 2010–2025 moves the committed value only within [\$1,838.32, \$1,854.65]. A Prais–Winsten AR(1) error model or a dollar-loss (level-form) objective moves it within [\$1,744, \$2,075], and re-fitting the slope while holding the intercept at the maintained β_A moves it to \$1,928 (+4.4 percent). All sit inside the anchor range that the robustness grid probes. Moving the *start* of the window is the consequential choice: the start-year curve peaks near 1970 (\$2,240 at a 1969 start; very late starts are weakly identified as the identifying variation shrinks), and the 1970-start estimate (\$2,177.24) is reported as a separately-labeled scenario in Section 5.4, never pooled with the end-year band. Formal time-series diagnostics (unit-root and cointegration tests, and difference/error-correction re-estimates) are in Appendix E.

²⁰The anchor regression's fitted intercept $\hat{\omega}_A$ is a parameter of the anchor category (food and beverages purchased for off-premises consumption) and never enters the calibration. That it lands near the inverted $\beta_A = 0.0250$ is a coherence remark between the time-series and cross-section routes, not a test. The two are estimates for different consumption categories, and no formal comparison is implied.

also implies a second, independent route to σ . With the committed position confined to food, services spending is entirely discretionary, so the model predicts $\tilde{\beta}_S^c = \beta_S \sigma^c$: dividing each economy's observed services share by the common β_S recovers σ^c , using no food-price data at all. Comparing this reading with the baseline calibration is a check on moments never targeted, and the result has two parts. The two routes agree in how committedness moves with GDP per capita (correlation 0.74, rank correlation 0.83): both say committedness is large where GDP per capita is low. At a given level of GDP per capita, they are essentially uncorrelated (partialling out log GDP per capita, the correlation is 0.035), so the agreement concerns the pattern, not a country-by-country validation. The same logic exposes where the common-preference-share restriction strains. Applied to manufacturing, it predicts $\tilde{\beta}_M^c = \beta_M \sigma^c$, yet that facet of the check fails (correlation -0.19), and the manufacturing-to-services expenditure ratio, which the model holds constant at 0.142, has a cross-country median of 0.206 and range [0.084, 0.550]. Appendix F turns the services-share reading into an explicit calibration bound.

What holding $\bar{c}_A$ common assumes, and its cost. Common preferences are the paper's transparency choice. Country-specific preference parameters could absorb culture, norms, home production, and measurement error wholesale, leaving the cross-country comparison uninterpretable. Variation on the production side, by contrast, is measured. However, the choice has a cost. When a common-preference demand system meets expenditure data it cannot fully fit, the unmodeled heterogeneity does not vanish. It loads into the calibrated committed shares, with a systematic sign: the model's implied food shares overshoot the observed ones for most included economies, and by the most where GDP per capita is lowest. In the poorest economies, the anchored committed bundle can cost more than everything households are observed to spend on food (the affordability failures below are the extreme form), so the poor-tail committed shares inherit an upward tilt from the linearity of the Engel curve being extrapolated.

I respond with two reporting conventions rather than by adjusting the calibration. First, the baseline is read throughout as a price-anchored upper bound: its poor-country committed shares carry the upward tilt just documented, so the magnitudes built on them are treated as upper bounds, not point estimates. Second, Appendix F quantifies the other side with two calibration bounds: one enforces the model's own affordability constraint economy by economy, and the other anchors committedness on each economy's own expenditure structure. The paper's central conclusions survive under both, though with smaller magnitudes. Two clarifications discipline the word *bounds*. They are bounds on the calibration within the maintained common-preference design, its most- and least-committed feasible states, and not identified-set bounds on preference parameters. And the corridor's lower edge is disciplined by the data rather than free. With a common preference share, committedness is what reconciles the observed Engel gap: observed agricultural expenditure shares have a poorest-quartile median of 0.228 against 0.069 in the richest quartile. Driving committedness toward zero would require the preference share to match the poor-country shares, 9.2 times the calibrated 0.0250, which would abandon the common-preference design rather than recalibrate it.

The sample, and where it fails. Imposing a common committed quantity on the poorest economies is demanding, and I report where it fails. At the calibrated parameters, 6 of the 80 economies (Bangladesh, Cambodia, Cameroon, the Democratic Republic of the Congo, Myanmar, and Senegal) have a committed food bill that meets or exceeds total household spending ($u_A^c \geq 1$): the anchored subsistence bundle is not affordable there at ICP prices. These economies are excluded from the baseline experiments and reported at the affordability boundary (Appendix F). The baseline sample is the remaining $n = 74$. How far the calibration extrapolates beyond the anchor’s support can itself be measured. The US committed share never leaves $[0.05, 0.22]$ over the estimation window, while 21 of the 74 included economies sit above that range (the poorest-quartile median a full 2.2 support-widths beyond it, the furthest included economy 3.8), and the anchored bundle can exceed everything households are observed to spend on food and alcohol, in economies as large as China and India (Appendix F). A direct Engel-curve fit across countries likewise yields a committedness gradient flatter than the anchored calibration’s (slope 0.30 in the model’s own units). The poor-tail committed shares, and every level built on them, are upper bounds, as stated above. Cross-country orderings are robust under this reading (the welfare-elasticity ranking carries a rank correlation of 0.98 with the capped state and 0.85 with the share-anchored alternative of Appendix F), while levels at the extreme tail are not.²¹

5 Quantitative Results

This section quantifies the theoretical insights developed in Section 3. The main exercises in this section are as follows: I hit every economy with the same negative sectoral productivity shock and compare the GDP response, the welfare response, and the practitioner’s benchmark built on the observed Domar weight, across countries. The benchmark shock is a 10 percent (precisely speaking, a 10-log-point) decline in the productivity of one sector, one sector at a time.²² Throughout, income quartiles are quartiles of 2017 GDP per capita (PPP) across the 80 table economies. Quartile statistics are computed on the included baseline sample ($n = 74$): the six excluded affordability-boundary economies all sit in the poorest quartile (Section 4.2). Means and medians are reported separately, with the basis labeled.

The next three subsections ask three questions of this one experiment. Section 5.1 compares the three measures economy by economy and establishes the paper’s measurement result: the

²¹The affordability census and the upper-bound disclosures matter because the level of the anchored committed value is the quantification’s single most consequential measurement choice. Section 5.4 spans the headline results across anchor variants, and the calibration bounds of Appendix F report the least-committed states the maintained design admits.

²²Because first-order responses are exact elasticities (Section 3), the shock size is a reporting convention. The second-order corrections that a 10 percent shock activates are small away from the subsistence corner, and Section 5.4 quantifies them.

observed Domar weight overstates the GDP response and understates the welfare response, and both discrepancies are largest in poor economies. Section 5.2 compares the same responses across rich and poor countries and establishes the paper’s development result: the same shock produces a mildly sloped GDP gradient but a welfare gradient an order of magnitude steeper. Section 5.3 asks where these patterns come from, decomposing each economy’s response into the contributions of subsistence demand, the production network, and their interaction. Section 5.4 collects robustness and two complementary results.

5.1 The Three Response Measures: GDP, Welfare, and the Observed Domar Weight

This subsection compares the three measures economy by economy: what each one says about the same shock, and where the observed Domar weight fails as a summary of the other two.

Figure 1 displays, for each economy, the three objects that Section 3 distinguishes for agriculture: the GDP elasticity λ_A^h , the observed Domar weight λ_A , and the welfare elasticity λ_A/σ , the observed Domar weight scaled by the calibrated multiplier.²³ The figure’s message is the section’s central pattern. In rich countries, the three measures nearly coincide, and the observed Domar weight is an adequate summary of either response. In poor countries, the three separate by more than an order of magnitude. The GDP elasticity remains modest while the welfare elasticity climbs to many times the observed Domar weight.

Three features of Figure 1 are crucial for the results that follow. First, the *observed Domar weight* falls with GDP per capita: quartile medians of 0.406 in the poorest quartile against 0.12 in the richest. This shows that agriculture occupies a far larger place in poor economies’ production and sales. Second, the *GDP elasticity* is far smaller and only mildly decreasing: quartile medians run from 0.086 to 0.058. Poor countries do lose more GDP from the same agricultural shock, but only modestly more, far less than their observed Domar weights suggest. Third, the *welfare elasticity*, the observed Domar weight amplified by $1/\sigma$, is far larger and falls steeply with GDP per capita: quartile medians run from 2.86 in the poorest quartile to 0.151 in the richest (on the model-consistent basis).

The mild slope of the GDP elasticity, and its exceptions, have an instructive anatomy. The decomposition $\lambda_A^h = \beta_A \ell_{AA} + \beta_M \ell_{MA} + \beta_S \ell_{SA}$ splits the GDP elasticity by the final good through which the shock reaches households. ℓ_{jA} is the agricultural cost content of good j ’s supply chain (the percent by which good j ’s price rises when agricultural productivity falls by one percent), and each term weights that content by the preference share spent on good j . The direct food channel $\beta_A \ell_{AA}$ is essentially flat across the quartiles (medians 0.0363 to 0.0348).

²³Figure 1 evaluates the welfare formula of Theorem 3 on each economy’s *observed* Domar weight. The counterfactual exercises of Section 5.3, and the welfare series and ratios of Figure 2, instead use the model-consistent elasticity, the closed-form solution of equation (21). The two differ only through the model’s Engel-curve fit, and the difference is conservative at the poor tail.

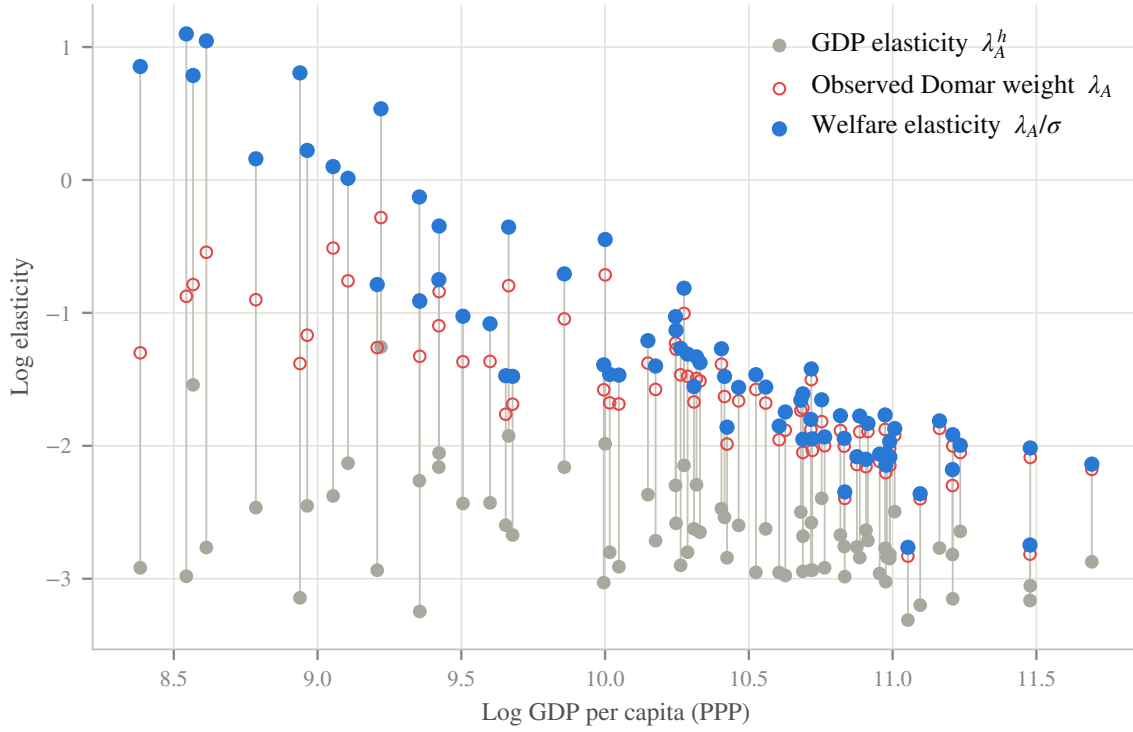


Figure 1: The three response measures: the agricultural elasticities of GDP and welfare, and the observed Domar weight

Notes: Each vertical segment spans the three measures for one economy in the included baseline sample ($n = 74$). Gray: the GDP elasticity λ_A^h (Theorem 2). Red open circles: the observed Domar weight λ_A (sales of the agriculture sector over GDP, 2017). Blue: the welfare elasticity λ_A/σ (Theorem 3: the percentage response of above-subsistence consumption), the observed Domar weight scaled by the calibrated multiplier. Both axes are in natural logarithms.

Households' food purchases give poor and rich countries nearly the same GDP exposure. Virtually the entire gradient comes instead from the indirect channels. Poor countries' services and manufacturing supply chains embed more agriculture, and the services term alone falls from 0.0419 to 0.0190 across the quartiles. The same decomposition explains the bottom tail, where composition matters more than the level of development. Agro-processing economies, whose production chains are strongly food-intermediated, carry the largest GDP elasticities in the sample (Vietnam at 0.285), while resource-enclave economies, where food is consumed but barely enters the cost structure of the rest of the economy, carry some of the smallest (0.043 for Angola). This is why Angola's GDP exposure and its welfare exposure sit at opposite extremes. The GDP elasticity runs through costs, while the welfare elasticity runs through the committed budget. These cross-country differences in network structure are themselves part of the paper's story: Section 5.3 returns to them, showing how the network's changing structure along the development path reshapes what shocks do to welfare.

Figure 2 converts the two elasticities into responses to the common 10 percent shock to each sector and adds the observed Domar weight, the object practitioners would actually use as a

response prediction. Section 3.2 concluded that, wherever the net committed position is positive, the observed Domar weight understates the welfare response of every sectoral shock, and that it overstates the GDP response of shocks to the necessity and to the industries supplying its chain (Theorem 3; Corollary 1). The three panels, one per shocked sector on a common scale, show both discrepancies in the data. Focus first on the agriculture panel. The observed-Domar-weight prediction sits *between* the two true responses for all but 2 of the 74 economies (and only marginally there): close enough to each to be mistaken for either among the richest economies, and misleading about both among the poorest. Against welfare, the observed Domar weight *understates*: the ratio of the welfare response to the observed-Domar-weight prediction has a poorest-quartile median of 5.41 and declines to 1.19 in the richest quartile. Against GDP, the observed Domar weight *overstates* for every single included economy: the ratio of the GDP response to the observed-Domar-weight prediction is below one throughout, with quartile medians rising from 0.20 (poorest) to 0.47 (richest). Importantly, both discrepancies are largest in the poorest countries, because both are made by the same object: the committed content of poor economies' food sales. It inflates the observed Domar weight without entering the GDP elasticity, and it enters the welfare response scaled up by the multiplier $1/\sigma$.

The manufacturing and services panels complete the picture and can be read quickly. The committed surcharge that separates welfare from GDP is routed by the network to the sectors feeding the food chain. At the poorest-quartile median, the welfare response exceeds the GDP response by a factor of 22.66 for the agricultural shock, against only 1.69 for manufacturing and 1.34 for services. *Which* shock is devastating, and *where*, is the pattern $(u_A/\sigma) \ell_{Aj}$ of Example 2. The systematic pattern of these responses across countries is the subject of the next subsection.

5.2 The Same Shock across Poor and Rich Economies: GDP versus Welfare

The previous subsection compared the three measures and, along the way, displayed their patterns across countries. This subsection zooms in on the cross-country dimension itself. The question is the paper's motivating one: how much more does the same shock cost a poor country than a rich one? The statistics of interest below are quartile summaries of the same responses in the agriculture panel of Figure 2. On the GDP side, the response is modestly decreasing in GDP per capita: quartile means fall by a factor of 1.76 from the poorest to the richest quartile (medians, 1.46). On the welfare side, the same shock produces losses 26.3 times larger in the poorest quartile than in the richest (means; medians, 19.0). The same shock, the same economies, the same accounting date: a gently sloped GDP gradient but a welfare gradient an order of magnitude steeper. At the country level, the contrast is starker still. The welfare cost of the shock is 2.3 times its GDP cost in the US, 6.7 times in Mexico, and 30.2 times in India. Nor is the contrast an artifact of the baseline calibration. Even under the conservative alternative calibrations of Appendix F, which push poor-country committedness down as far as the

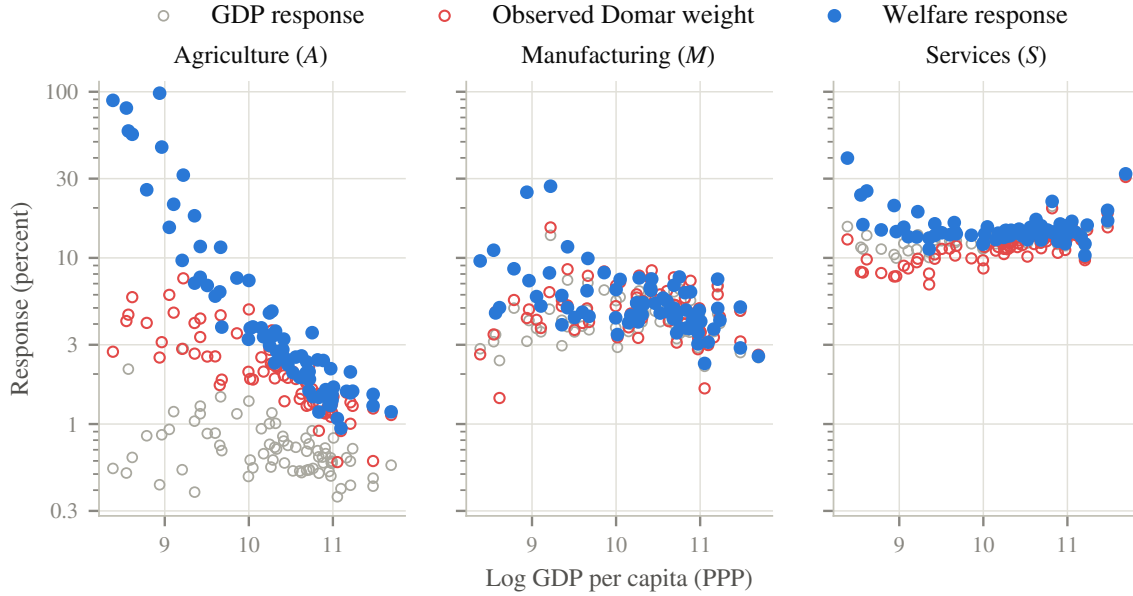


Figure 2: Responses to a common 10 percent sectoral shock, by shocked sector: welfare, GDP, and the observed Domar weight

Notes: First-order responses to a 10 percent (10-log-point) productivity decline in the sector named in each panel, for the included sample ($n = 74$). Responses are measured in log points (one hundred times the absolute log change) and are read as percent under the small-change approximation. At the largest responses (the poorest economies in the agriculture panel), the approximation is loose, and the log-point values are the ones to read. Appendix C compares the first-order and exact responses to the agricultural shock. The horizontal axis is the log of GDP per capita, while the vertical axis is logarithmically spaced and common across panels. Blue: the welfare response (model-consistent basis: see footnote 23). Red open: the observed Domar weight's prediction ($0.10 \lambda_j$; by Section 3.4, also the pointwise response of chained real GDP). Gray open: the GDP response.

maintained design allows, the welfare gradient remains 3.7 to 4.0 times the GDP gradient.²⁴

A pair of economies makes the sufficient-statistic point concrete. Poland and Angola carry observed agricultural Domar weights of 0.2501 and 0.2513, numerically nearly equal, at incomes that differ by a factor of 4.3. The equality is no accident of measurement: Poland's large food-processing industry makes its agri-food sales unusually high for its income level, while Angola's oil-dominated GDP compresses its share to well below its poorest-quartile peers (median 0.406). The two composition extremes meet at the same sales ratio. Any accounting built on the observed Domar weight assigns them the same exposure to agricultural shocks. Their calibrated welfare responses differ by a factor of 7.93 on the conservative σ -ratio basis. The observed Domar weight is not a small-sample casualty here. It is doing what Corollary 1 and Theorem 3 say it does: recording average expenditure, not marginal response, and missing the

²⁴The observed Domar weight supplies the taxonomy's third reading. As the pointwise response coefficient of chained real GDP and of the endowment metric (Section 3.4), its own quartile gradient is a factor of 3.4 at medians, between the paper's two measures, the ordering the taxonomy requires.

multiplier.

There are two remarks on the cross-country differences. First, per Section 4.2, the poor-tail committed shares are upper bounds, so the levels of the poorest-quartile responses are upper bounds, while the cross-country ordering is robust. Second, the 6 affordability-boundary economies (Bangladesh, Cambodia, Cameroon, the Democratic Republic of the Congo, Myanmar, and Senegal) are excluded from all of these statistics. The gradient is computed on economies where the interior theory applies, and the six are precisely the economies the calibration judges most subsistence-exposed, so the reported statistics omit the calibration’s own steepest cases, not its mildest.

5.3 The Role of Production Networks: A Counterfactual Decomposition

The results so far mix the model’s two mechanisms: subsistence demand (the committed position $\bar{c}_A > 0$) and input–output propagation (the Leontief inverse L). This subsection separates them and answers the attribution question directly: how much of poor countries’ vulnerability is subsistence demand, how much is the production network, and how does the answer change along the development path?

The closed-form solution of Section 3.3 makes the attribution exact, in two complementary readings of the same object. First, because it makes the welfare elasticity an explicit function of (L, β, u_A, σ) , each mechanism can be switched off. I evaluate every economy’s elasticity in four counterfactual worlds (full, homothetic, no-network, and neither), and Figure 3 reports the levels. Second, equation (22) in Example 2 decomposes the full welfare elasticity into four additive terms: the neither world’s level (direct final demand), the increment from adding the network alone (network only), the increment from adding the non-homotheticity alone (non-homotheticity only), and what only the pair produces (the interaction). Figure 4 reports the four terms as shares of each economy’s response. The two readings answer different questions. The former asks what each economy’s response would be with a mechanism removed, and so identifies which mechanism creates the cross-country differences, while the latter asks how much of the actual response each mechanism carries, and so tracks what the network does at each level of development.

Four worlds. The four worlds are: the *full* model; the *homothetic world*, which keeps the measured network but sets $\bar{c}_A = 0$ (there welfare and GDP coincide, and the elasticity is λ_A^h); the *no-network world*, which keeps the calibrated demand system but removes all intermediate inputs ($\Omega = O$, so that $L = I$), the production structure of the “horizontal economy” that the production network literature uses as the benchmark without input–output linkages (e.g., Baqaee & Farhi, 2019; Baqaee & Rubbo, 2023); and the *neither* world (both switched off), whose elasticity is the direct final-demand share β_A . Throughout, the counterfactuals follow the convention standard in that literature: the calibrated demand-side statistics (β, u_A, σ) are held at their measured values and only the propagation operator is swapped, so the four worlds decompose

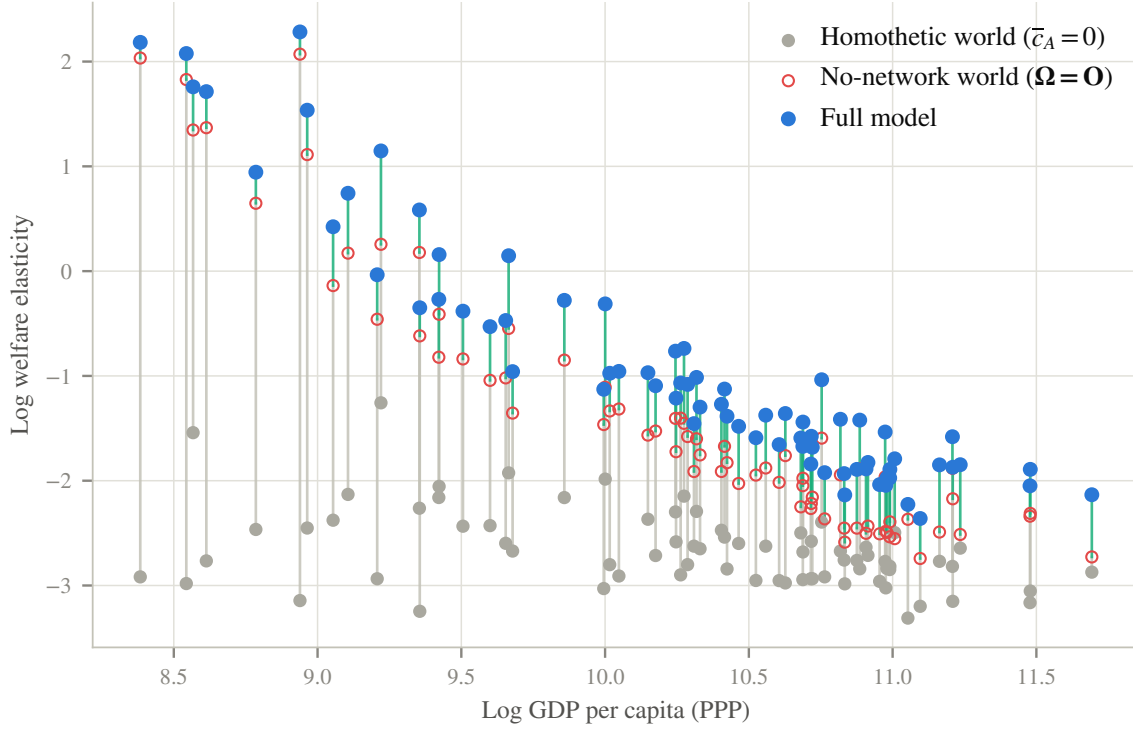


Figure 3: Counterfactual worlds: the welfare elasticity in the full model, the no-network world, and the homothetic world

Notes: Welfare elasticities to the agricultural shock per economy ($n = 74$; both axes in natural logarithms). Blue: the full model. Red open: the no-network world ($\Omega = \mathbf{O}$, so that $\mathbf{L} = \mathbf{I}$). Gray: the homothetic world ($\bar{c}_A = 0$; welfare $\equiv$ GDP). Light gray connectors span the two single-mechanism worlds; green connectors span the network's total contribution (no-network world to full model).

each economy's response operator at its measured state. They are not re-solved equilibria.²⁵

Figure 3 plots three of the four worlds for each economy, omitting the neither world, which is the common constant $\beta_A = 0.0250$. The four terms of equation (22) read as follows: *network only* is the homothetic world's elasticity in excess of the base ($\lambda_A^h - \beta_A$: what the network contributes with the non-homotheticity switched off); *non-homotheticity only* is the no-network world's excess (u_A/σ : the committed position itself, with no network to propagate it); and the *interaction* is what only the pair produces ($(u_A/\sigma)(\ell_{AA} - 1)$: the network's own-loop multiplying the committed position). The terms are exact and sum to the full welfare elasticity economy by economy; and because β_A is the common anchored constant, every cross-country pattern in Figure 4 is carried by the country-specific $(\mathbf{L}, u_A, \sigma)$. What varies across economies in the base band is only its share of a country-specific total, not the number β_A itself.

Levels: the gradient is demand-made; the network multiplies. Figure 3 delivers the attribution in levels. The full welfare elasticity falls from a poorest-quartile median of 2.86 to a richest-

²⁵The convention is also the only one the sufficient statistic identifies: re-solving counterfactual equilibria would require productivity levels, which $(\mathbf{L}, \beta, \mathbf{u})$ deliberately never uses.

quartile median of 0.151, a factor of 19. Strip out the entire production network and the cross-country differences survive essentially intact: the no-network world falls from 1.60 to 0.085, a factor of 19. Strip out only the non-homotheticity instead and the cross-country differences disappear: the homothetic world's factor is 1.5, the only mildly decreasing GDP elasticity of Figure 1. These results imply that the demand side determines which countries are vulnerable: subsistence committedness by itself accounts for most of the cross-country differences, while the network by itself plays a more limited role. What the network does is different and twofold. First, it multiplies: the full response exceeds the no-network response by a per-country median factor of 1.6, and the amplification is remarkably uniform across countries (quartile medians 1.5 at the bottom, 1.6 at the top). Development flattens the welfare-cost curve by pulling economies off the committed floor, not by rewiring away the network's amplification. Second, it decides which shocks, and how fast: which sectors' shocks carry the committed surcharge (Figure 2), and how fast an adverse path approaches the subsistence corner. Both margins are network-made: the committed surcharge reaches sector j in proportion to the food chain's exposure ℓ_{Aj} , concentrating the excess welfare losses in agricultural and food-chain shocks (Figure 2), and the distance to an economy's subsistence threshold is its committed headroom divided by the own-multiplier ℓ_{AA} (Appendix C), so the same demand state sits closer to the corner in a more food-intermediated economy. The network amplifies every economy's response without creating the differences between them.

The rotation: a constant network share whose job changes. Figure 4 stacks the four terms of equation (22) as shares of each economy's full welfare elasticity, for all 74 economies ranked by GDP per capita. Three findings emerge. First, the network's total share (the network-only and interaction bands together) is a near-constant third of the welfare response at every income level: quartile medians of 33.6, 40.1, 38.7, and 38.4 percent from poorest to richest. The network rarely carries even half of the welfare response, and never vanishes from it. Second, within that constant third, the composition inverts along the development path. At the poorest-quartile median, the interaction carries 29.7 percent of the full response against 2.6 percent for the network-only term. Among the poorest economies, the network matters almost entirely because it multiplies subsistence exposure. At the richest-quartile median, the ordering flips (21.8 percent network-only against 17.3 percent interaction). Among the richest, the network matters as the cost-transmission channel of the GDP response. Hence, poor countries' networks amplify needs, while rich countries' networks transmit costs. Third, the mechanism of the rotation is a race of decay rates. From the poorest to the richest quartile, the demand-side statistic u_A/σ collapses by a factor of about 26 (from 1.58 to 0.060 at the quartile medians) while the supply-side term $\lambda_A^h - \beta_A$ shrinks only by a factor of 1.8. The committed channel simply vanishes faster than the cost channel.

There are two remarks about the rotation. First, it is a statement about composition, not about magnitudes. Every term but the common base β_A , which is constant in level across economies, falls with development, and the network's absolute contribution drops from 1.07

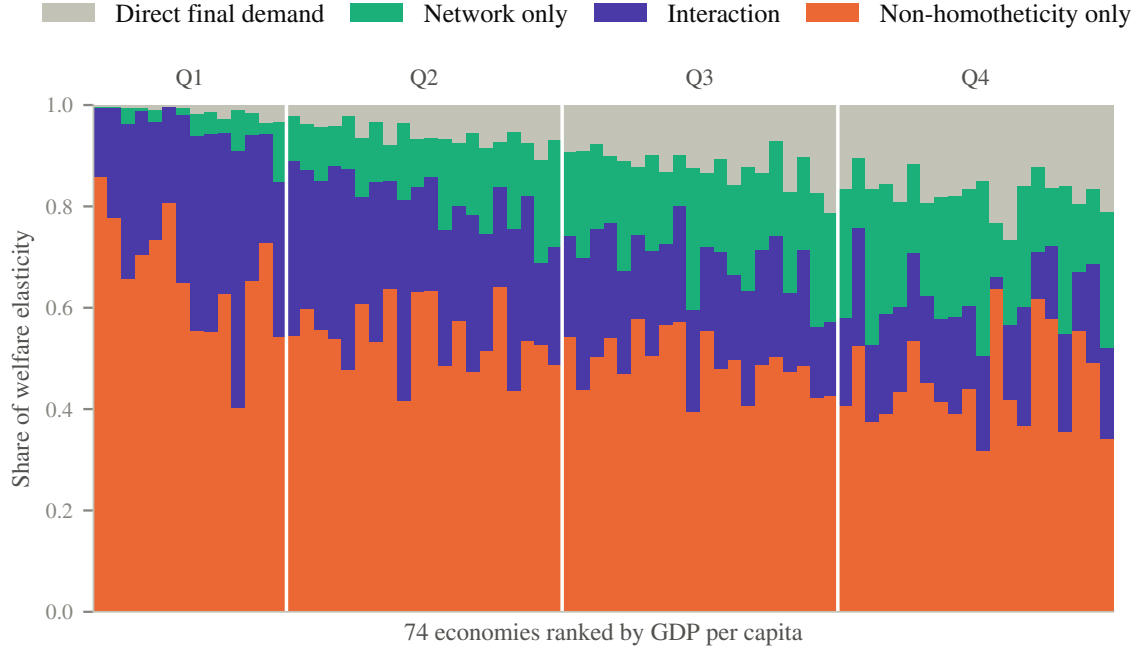


Figure 4: The rotation: the four components of the welfare elasticity, as shares, along the development path

Notes: For each included economy (one bar; $n = 74$, ranked by 2017 GDP per capita (PPP); quartiles separated), the four terms of the decomposition (22) as shares of the full welfare elasticity, under the convention of Figure 3: direct final demand (β_A , common across economies in level), network only ($\lambda_A^h - \beta_A$), non-homotheticity only (u_A/σ), and the interaction ($(u_A/\sigma)(\ell_{AA} - 1)$). Shares, not levels: every component but the constant base β_A falls with GDP per capita in levels (see text).

to 0.058 elasticity units at the quartile medians (a factor of 19) even as its share holds. Second, the denominator matters. On the GDP side, the network-carried share of λ_A^h falls with GDP per capita (from 71 to 57 percent), because poor economies' small GDP responses are more network-intermediated than rich economies'. The two statements have different denominators, and both are reported.

The measured counterpart: agriculture's intermediate-input basket modernizes. The rotation is not a model artifact, and its measured counterpart sits in the raw input–output tables. Table 1 reports the cost structure of *primary* agriculture, defined as ISIC A01–A03,²⁶ by income quartile. Purchased cross-sector inputs rise steeply with development. The manufactured-inputs share (fertilizer, agro-chemicals, fuel, machinery services) roughly doubles between the poorest and middle-income quartiles, and the services-inputs share (trade, finance, veterinary and agronomic services) more than triples from the poorest to the richest quartile, the steepest

²⁶Primary agriculture is the right unit here. The modernization facts concern farming's purchased inputs, and the literature cited below measures input intensity on farms. In the integrated sector of the main text, food processing, an intermediate-heavy manufacturing activity in rich and poor economies alike, would dominate the cost shares and obscure the gradient. The primary rows are native to the ICIO source tables (Appendix E).

Table 1: The modernization of agriculture’s intermediate-input basket along the development path

Income quartile	Ω_{AM}	Ω_{AS}	Ω_{AA}	α_A	ℓ_{AA}
Q1 (median)	0.088	0.070	0.125	0.678	1.165
Q2 (median)	0.160	0.144	0.153	0.513	1.210
Q3 (median)	0.214	0.178	0.142	0.441	1.184
Q4 (median)	0.189	0.225	0.158	0.451	1.200
Rank corr. with GDP p.c.	0.39	0.67	0.02	-0.55	-0.02

Notes: Cost shares of *primary* agriculture (ISIC A01–A03; here, and only here, the subscript A refers to primary agriculture), by income quartile (medians over included economies; $n = 74$): purchased inputs from manufacturing (Ω_{AM}) and services (Ω_{AS}), own-sector inputs (Ω_{AA}), the value-added share (α_A), and the own Leontief multiplier (ℓ_{AA}). The last row reports rank correlations with GDP per capita across economies.

input-share gradient in the table. The mirror image is the value-added share. Poor-country agriculture is the value-added-intensive activity: labor and land, few purchased inputs of any kind. Meanwhile, the *own*-sector share is flat across the quartiles, and so is the own multiplier ℓ_{AA} of the main text’s agriculture sector (quartile medians 1.45 to 1.39). The modern-inputs revolution runs through *other* sectors’ columns of the Leontief inverse, not through the own-loop. This confirms the role of intermediate inputs highlighted in the literature on cross-country agricultural productivity differences (e.g., Restuccia et al., 2008; Donovan, 2021; Boppart et al., 2026). In particular, Boppart et al. (2026) document input intensification as the dominant margin behind agricultural productivity differences, concluding that productivity gains in the sectors producing agriculture’s inputs are a lever for agricultural development, a transmission statement that the columns ℓ_{Aj} formalize. The rotation is this modernization seen from the welfare side: as the intermediate-input basket deepens and diversifies, agricultural shocks reach welfare less through the committed budget and more through input-cost transmission. This reading interprets a 2017 cross-section as a development path.

5.4 Robustness and Complements

Robustness. Appendix F reports a series of robustness checks along four main dimensions: the subsistence anchor (the borrowed estimate without re-fitting, alternative estimation windows, committed quantities scaled down by a quarter or a half, and higher preference shares for food); the measurement conventions (the 2021 tables and prices instead of 2017, domestic-only input–output flows, the treatment of trade and transport margins, alternative international price levels, a food category including alcoholic beverages, and separating primary agriculture from food processing); the calibration bounds of Section 4.2; and the demand configuration itself

(household-only budgets). Each variant re-computes the affordability census, so the excluded set adjusts with the calibration. Three conclusions summarize these checks. First, the paper’s central contrast is never overturned. The poorest-to-richest welfare gradient stays above a factor of 9.3 in every variant, against the GDP gradient’s factor of 1.46. Second, the single most consequential input is the level of the anchored committed value, not anything about the network. The conservative calibration bounds compress the welfare gradient to a factor of 5.4 to 5.9, still several times the GDP gradient, and it is the poorest-tail levels that move with that value, which is why Section 4.2 reads them as upper bounds. The observed Domar weight understates the welfare response for the large majority of economies under every calibration state. Third, the one variant that changes the sample itself is the start-year anchor scenario, under which the affordability census grows to 9 economies. It is reported separately and never pooled.

The welfare price of volatility (complement). The same apparatus extends from one-off shocks to recurring sectoral *volatility*. To second order, expected log-welfare losses from mean-zero sectoral fluctuations are governed by the committed exposure $(u_A/\sigma)\ell_{AA}$, so the model converts observed sectoral TFP volatilities into a welfare price of volatility per economy. Evaluated at common sectoral shock moments, that price is 89 times larger at the poorest-quartile median than at the richest (an order of magnitude steeper than the multiplier gradient itself: the price is second order, loading on the committed exposure squared per unit of committed share, $[(u_A/\sigma)\ell_{AA}]^2/u_A$), and its composition flips along the development path, from basket-price risk in rich economies to subsistence risk in poor ones. Margin and shock-scale variants dampen the price at the poorest-quartile median by factors of 0.55 and 0.69. This complements the paper’s results, and Appendix C develops it fully.

How much does second order add? (complement). GDP needs no correction. Its response is exactly linear at all orders (Theorem 2). For welfare, the second-order correction to the 10 percent agricultural shock is a rounding error in rich economies (a second-to-first-order ratio of 0.044 at the richest-quartile median) and material only near the subsistence corner (0.21 at the poorest-quartile median). Away from the corner, the first-order formulas scaled by $1/\sigma$ are an accurate summary, which is what makes the sufficient-statistic apparatus usable. The full second-order layer (curvature, the loss–gain asymmetry, and each economy’s distance to its subsistence threshold) is characterized and quantified in Appendix C.

6 Conclusion

The same sectoral shock can devastate a poor country and leave a rich one largely unaffected. This paper explains this contrast in a canonical multisector model with input–output linkages and one deviation: the final-demand aggregator is non-homothetic. The model admits a closed-form solution and delivers a taxonomy. Hulten’s theorem fails with respect to GDP but holds

for welfare measures up to a country-specific scalar, the inverse of the discretionary share. The observed Domar weight, the practitioner's sufficient statistic, can therefore mismeasure both responses. In the leading case in which food is a single necessity, it overstates the GDP impact of agricultural shocks and understates their welfare impact.

Quantitatively, calibrating the model to 80 economies with non-homotheticity terms anchored to a six-decade US expenditure series, the same 10 percent agricultural shock produces GDP losses only modestly larger in poor countries, with quartile medians a factor of 1.46 apart, but welfare losses 19.0 times larger across the same divide. Subsistence demand determines which countries are vulnerable to sectoral shocks, while the production network multiplies every economy's losses and decides which sectors' shocks are devastating. The network's role rotates along the development path, from amplifying subsistence needs in poor countries to transmitting input costs in rich ones. GDP and welfare answer different questions in poor economies, and the difference is largest where sectoral shocks hurt the most.

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A Proofs and Derivations

This appendix collects all derivations and proofs for Sections 2 and 3.

A.1 Household and Firm Optimization

Household. Maximizing (1) subject to (2) yields the linear expenditure system:

$$p_i c_i = p_i \bar{c}_i + \beta_i K, \quad K := w - \sum_{k=1}^n p_k \bar{c}_k, \quad (\text{A.1})$$

where K is the household's discretionary income (the *supernumerary income* of the linear-expenditure-system tradition). The household first purchases its committed bundle and then spends the remainder in the fixed proportions β . Substituting (A.1) into (1) and using $\kappa_i = \beta_i^{-\beta_i}$ gives an indirect utility function:

$$C = \prod_{i=1}^n \kappa_i \left(\frac{\beta_i K}{p_i} \right)^{\beta_i} = \frac{K}{P}, \quad \text{where} \quad P := \prod_{i=1}^n p_i^{\beta_i}. \quad (\text{A.2})$$

Welfare is discretionary income deflated by the price index. Dividing (A.1) by w and using the definitions (6) delivers the expenditure shares (8), $\tilde{\beta}_i = u_i + \beta_i \sigma$, together with $K = \sigma w$.

Firms. The first-order conditions of the maximization problem of (4) subject to (3) are: $wh_i = \alpha_i p_i y_i$ and $p_j x_{ij} = a_{ij} p_i y_i$. Substituting them into (3) and using the normalization ζ_i yields:

$$\log \left(\frac{p_i}{w} \right) = -\log A_i + \sum_{j=1}^n a_{ij} \log \left(\frac{p_j}{w} \right). \quad (\text{A.3})$$

A.2 Proof of Proposition 1

(i) *Prices.* Stacking (A.3) over i and solving the linear system gives $\log \mathbf{q} = -\mathbf{L} \log \mathbf{A}$, which is (7). Note that relative prices are determined by the supply side alone.

(ii) *Shares.* Established in (A.1)–(A.2) above.

(iii) *Domar weights.* In the goods market clearing condition (5), multiply by p_i and substitute the household demand $p_i c_i = \tilde{\beta}_i w$ and the input demands $p_i x_{ji} = a_{ji} p_j y_j$ to get:

$$p_i y_i = \tilde{\beta}_i w + \sum_{j=1}^n a_{ji} p_j y_j.$$

Dividing by $\text{GDP} = w$ and solving the resulting linear system $\boldsymbol{\lambda} = \tilde{\boldsymbol{\beta}} + \boldsymbol{\Omega}^\top \boldsymbol{\lambda}$ gives $\boldsymbol{\lambda} = \mathbf{L}^\top \tilde{\boldsymbol{\beta}}$. Substituting $\tilde{\boldsymbol{\beta}} = \sigma \boldsymbol{\beta} + \mathbf{u}$ yields $\boldsymbol{\lambda} = \sigma \boldsymbol{\lambda}^h + \mathbf{L}^\top \mathbf{u}$, and the wedge form follows from $1 - \sigma = \sum_i u_i$:

$$\lambda_j - \lambda_j^h = \left(\mathbf{L}^\top \mathbf{u} \right)_j - (1 - \sigma) \lambda_j^h = \sum_{i=1}^n u_i \ell_{ij} - \sum_{i=1}^n u_i \lambda_j^h = \sum_{i=1}^n u_i \left(\ell_{ij} - \lambda_j^h \right).$$

(iv) *GDP and prices.* The numéraire imposes $0 = \sum_i \beta_i \log p_i = \sum_i \beta_i \log q_i + \log w$, so

$$\log \text{GDP} = \log w = - \sum_{i=1}^n \beta_i \log q_i = \sum_{i=1}^n \beta_i \sum_{j=1}^n \ell_{ij} \log A_j = \sum_{j=1}^n \lambda_j^h \log A_j,$$

and $\log p_i = \log q_i + \log w$ delivers the price formula in (10).

(v) *Welfare.* From (A.2), $\mathcal{C} = K/P = \sigma w/1 = \sigma \cdot \text{GDP}$ under the numéraire.

Existence, uniqueness, and interiority. Given A , relative prices are uniquely pinned down by (i), the wage by the numéraire, and consumption by (A.1); Assumption 2 guarantees $K = \sigma w > 0$ and $c_i > 0$, so the household's problem has an interior solution and the first-order conditions used above are valid. Sales follow uniquely from (iii), outputs from $y_i = \lambda_i w / p_i > 0$, and inputs and labor from the firms' first-order conditions. Labor market clearing holds by construction: $\sum_i h_i = \sum_i \alpha_i \lambda_i = \alpha^\top L^\top \tilde{\beta} = (L\alpha)^\top \tilde{\beta} = \mathbf{1}^\top \tilde{\beta} = 1$, where $L\alpha = \mathbf{1}$ follows from $(I - \Omega)\mathbf{1} = \alpha$, and $\mathbf{1}^\top \tilde{\beta} = \sigma + \sum_i u_i = 1$. Every step is an equivalence, so the equilibrium is unique. $\square$

A.3 Proof of Proposition 2

Differentiating (7) gives (12). For (13): it follows from $u_k = \bar{c}_k q_k$ that

$$du_k = u_k d\log q_k = -u_k \sum_j \ell_{kj} d\log A_j,$$

and summing over k in $\sigma = 1 - \sum_k u_k$ gives

$$d\sigma = \sum_j \left(\sum_k u_k \ell_{kj} \right) d\log A_j = \sum_j m_j d\log A_j.$$

For (14): differentiate $\tilde{\beta}_i = \sigma \beta_i + u_i$ term by term. Adding up: $\sum_i (\beta_i m_j - u_i \ell_{ij}) = m_j - m_j = 0$ for every j . $\square$

A.4 Proof of Theorem 1

Differentiate the closed-form expression $\lambda_i = \sigma \lambda_i^h + \sum_k u_k \ell_{ki}$ using Proposition 2 to obtain:

$$d\lambda_i = \lambda_i^h d\sigma + \sum_k \ell_{ki} du_k = \sum_j \left(\lambda_i^h m_j - \sum_k u_k \ell_{kj} \ell_{ki} \right) d\log A_j,$$

which is the first display in (15). Substituting $m_j = \sum_k u_k \ell_{kj}$ and collecting the sum over k gives the second. (i): using $\alpha^\top L^\top \tilde{\beta} = 1$ and $\sum_i \alpha_i \ell_{ki} = (L\alpha)_k = 1$,

$$\sum_i \alpha_i d\lambda_i = \sum_j \sum_k u_k \ell_{kj} \left(\underbrace{\sum_i \alpha_i \lambda_i^h}_{=1} - \underbrace{\sum_i \alpha_i \ell_{ki}}_{=1} \right) d\log A_j = 0.$$

(ii) is immediate from (15) with $\mathbf{u} = \mathbf{0}$. Finally, the claim in the text that the response matrix is nonzero at every admissible state when $\bar{c} \neq \mathbf{0}$: since $\lambda = L^\top \tilde{\beta}$ with $L^\top$ invertible, the Jacobian

of λ vanishes identically at a state if and only if the Jacobian of $\tilde{\beta}$ does, i.e., if and only if $\beta_i m_j = u_i \ell_{ij}$ for all i, j . Suppose so. Since $\mathbf{u} \neq \mathbf{0}$, $\mathbf{m} = \mathbf{L}^\top \mathbf{u} \neq \mathbf{0}$; pick j^* with $m_{j^*} \neq 0$. Then for every i , $u_i \ell_{ij^*} = \beta_i m_{j^*} \neq 0$, so $u_i \neq 0$ for all i , and dividing, $\ell_{ij} = (\beta_i / u_i) m_j$ for all i, j : every row of $\mathbf{L}$ is proportional to $\mathbf{m}^\top$, so $\mathbf{L}$ has rank one, contradicting invertibility, since $n \geq 2$. $\square$

A.5 Proof of Lemma 1

Fix j . The entry ℓ_{kj} of $\mathbf{L} = \sum_{m \geq 0} \mathbf{\Omega}^m$ is the total weight of all walks from k to j , where a walk's weight is the product of the coefficients a along it. Decompose each walk from k to j (k arbitrary) at its first visit to j : every walk factors uniquely into a walk from k to j that does not visit j before its endpoint, followed by a walk from j to j . Writing f_{kj} for the total weight of the first kind (with $f_{jj} = 1$, the empty walk),

$$\ell_{kj} = f_{kj} \ell_{jj}. \quad (\text{A.4})$$

The first-passage weights satisfy, for $k \neq j$, the recursion $f_{kj} = a_{kj} + \sum_{r \neq j} a_{kr} f_{rj}$. Let $\bar{f} = \max_{k \neq j} f_{kj}$, attained at k^* . Then

$$\bar{f} \leq a_{k^*j} + \left(\sum_{r \neq j} a_{k^*r} \right) \bar{f} = a_{k^*j} + (1 - \alpha_{k^*} - a_{k^*j}) \bar{f},$$

so $\bar{f}(\alpha_{k^*} + a_{k^*j}) \leq a_{k^*j}$ and hence $\bar{f} \leq a_{k^*j} / (\alpha_{k^*} + a_{k^*j}) < 1$: no industry's first-passage weight to j reaches one, because every step leaks a positive labor share. Now take $i \neq j$. Peeling off the first step of every walk from i to j and applying (A.4),

$$\ell_{ij} = \sum_{k=1}^n a_{ik} \ell_{kj} = \left(\sum_{k=1}^n a_{ik} f_{kj} \right) \ell_{jj} \leq \left(\sum_{k=1}^n a_{ik} \right) \ell_{jj} = (1 - \alpha_i) \ell_{jj}.$$

Finally, since $\ell_{jj} \geq 1$ and $\beta_j < 1$,

$$\lambda_j^h = \beta_j \ell_{jj} + \sum_{i \neq j} \beta_i \ell_{ij} \leq \beta_j \ell_{jj} + \sum_{i \neq j} \beta_i (1 - \alpha_i) \ell_{jj} < \left(\sum_i \beta_i \right) \ell_{jj} = \ell_{jj},$$

where the last inequality is strict because $\alpha_i > 0$ and $\beta_i > 0$ for the (nonempty) set of $i \neq j$. $\square$

A.6 Proof of Corollary 1

The wedge identity (16) restates (9). For the knife edge: $\lambda = \lambda^h$ if and only if $\mathbf{L}^\top (\tilde{\beta} - \beta) = \mathbf{0}$, and since $\mathbf{L}^\top$ is invertible, if and only if $\tilde{\beta} = \beta$. By (8), $\tilde{\beta} - \beta = \mathbf{u} - (1 - \sigma)\beta$, which vanishes exactly when $\mathbf{u} = (\sum_k u_k) \beta$. For the single necessity, $\mathbf{u} = u_F \mathbf{e}_F$, so the wedge for the food shock is $\lambda_F - \lambda_F^h = u_F(\ell_{FF} - \lambda_F^h)$, strictly positive by Lemma 1 when $\beta_F < 1$ and $u_F > 0$. $\square$

A.7 Proof of Theorem 3

From $\log \mathcal{C} = \log \text{GDP} + \log \sigma$ (Proposition 1(v)), Theorem 2, and (13),

$$d \log \mathcal{C} = \sum_j \left(\lambda_j^h + \frac{m_j}{\sigma} \right) d \log A_j.$$

By the Domar decomposition (9), $m_j = (\mathbf{L}^\top \mathbf{u})_j = \lambda_j - \sigma \lambda_j^h$, so $\lambda_j^h + m_j/\sigma = \lambda_j/\sigma$, which is (18); and $\lambda_j/\sigma = (p_j y_j / \text{GDP}) / (\sigma) = p_j y_j / (\sigma \text{GDP})$, with $\sigma \text{GDP} = \text{GDP} - \sum_k p_k \bar{c}_k$ the market value of discretionary expenditure. Equation (19) is the same computation displayed by channels.

□

A.8 Proofs for Section 3.4

Fixed-weight deflators and chained GDP. For any weight vector $\mathbf{b}$ with $\sum_i b_i = 1$,

$$d \log w - \sum_i b_i d \log p_i = \sum_i b_i (d \log w - d \log p_i) = - \sum_i b_i d \log q_i = \sum_j \left(\mathbf{L}^\top \mathbf{b} \right)_j d \log A_j,$$

using (12). With constant $\mathbf{b}$ this is the differential of the level index $w / \prod_i p_i^{b_i}$, which proves (24); with the current weights $\mathbf{b} = \tilde{\boldsymbol{\beta}}(\mathbf{A})$ it defines the chained growth rate, and $\mathbf{L}^\top \tilde{\boldsymbol{\beta}} = \boldsymbol{\lambda}$ proves (25). Because $\boldsymbol{\lambda}$ is state-dependent, (25) is nonlinear in the shock. Because the field $\boldsymbol{\lambda}(\mathbf{A})$ has an asymmetric Jacobian (Proposition 3), it is generically not the differential of any level index, and its integral over a discrete shock depends on the path, as characterized in Appendix B.

The cancellation of the Domar-weight changes. Differentiating $\log \text{GDP} = \sum_i \lambda_i \log A_i + \sum_i u_i \log p_i$ (the income-side identity obtained from (9) and the numéraire) produces, alongside λ_j , terms in $d\lambda_i$ and in the revaluation of u_i . Substituting the closed-form expressions shows the $d\lambda$ terms cancel against the price-level terms identically, leaving the constant λ_j^h of Theorem 2. The computation makes explicit that no derivative of any Domar weight survives into a first-order aggregate.

The endowment metric. Scale the labor endowment to L units. The budget hyperplane becomes $\mathbf{q}^\top \mathbf{c} \leq L$, and the argument of (A.1)–(A.2) gives the planner's value $V(\mathbf{A}, L) = (L - \sum_k q_k \bar{c}_k) / \prod_k q_k^{\beta_k}$, so that

$$\left. \frac{\partial \log V}{\partial \log L} \right|_{L=1} = \frac{1}{\sigma}, \quad \frac{\partial \log V}{\partial \log A_j} = \frac{\lambda_j}{\sigma}.$$

Following Baqaee and Burstein (2023), define EV^M as the log endowment equivalent of a shock: the solution ϕ of $V(\mathbf{A}_0, e^\phi) = V(\mathbf{A}_1, 1)$. Differentiating this identity at $\mathbf{A}_1 = \mathbf{A}_0$ in an arbitrary shock direction,

$$\frac{1}{\sigma} d\phi = \sum_j \frac{\lambda_j}{\sigma} d \log A_j \quad \implies \quad d \text{EV}^M = \sum_j \lambda_j d \log A_j,$$

which is (26); the relation $d \log \mathcal{C} = d \text{EV}^M / \sigma$ restates Theorem 3.

□

B Alternative Measures: Path Dependence and Discrete Comparisons

This appendix completes Section 3.4. It characterizes the second-order behavior of chain-weighted real GDP, shows that its path dependence is governed by a closed-form object, quantifies that object on the calibrated sample, and records the discrete-comparison caveat for the endowment metric.

The chained index at second order. The chained growth rate (25) has state-dependent weights, and its curvature is exactly the response of the Domar weights in Theorem 1. Writing $J_{jm} := \partial \lambda_j / \partial \log A_m = \lambda_j^h m_m - \sum_k u_k \ell_{km} \ell_{kj}$, the discrete response to a shock vector Δ satisfies

$$\Delta \log Y^{\text{ch}} = \sum_{j=1}^n \lambda_j^0 \Delta_j + \frac{1}{2} \sum_{j=1}^n \sum_{m=1}^n J_{jm} \Delta_j \Delta_m + O(\|\Delta\|^3),$$

when the shocks are phased in proportionally, along the straight-line path from $\mathbf{0}$ to Δ in log productivities. For the paper's welfare and fixed-weight GDP measures, the corresponding expansions are exact at first order (Theorem 2) and given in closed form in Appendix C.

Path dependence in closed form. Because J is not symmetric, the integral that defines the chained index over discrete horizons depends on the path of shocks, not only on endpoints. The circulation is pinned down by the antisymmetric part of J .

Proposition 3 (The curl of the chained index). *Apply a shock ε to sector j and a shock δ to sector m , in the two possible orders. To second order, the difference between the chained-GDP responses along the two paths is*

$$\Delta \log Y_{(j \rightarrow m)}^{\text{ch}} - \Delta \log Y_{(m \rightarrow j)}^{\text{ch}} = \varepsilon \delta (J_{mj} - J_{jm}) + O(3) = \varepsilon \delta (\lambda_m^h m_j - \lambda_j^h m_m) + O(3). \quad (\text{B.1})$$

The antisymmetric part vanishes for all sector pairs at a state if and only if $\mathbf{m} \propto \lambda^h$, the same knife-edge at which the observed Domar weights coincide with the homothetic-benchmark weights (Corollary 1).

Proof. A second-order expansion of the path integral of (25) along each ordering gives $\lambda_j^0 \varepsilon + \lambda_m^0 \delta + \frac{1}{2} J_{jj} \varepsilon^2 + \frac{1}{2} J_{mm} \delta^2 + J_{mj} \varepsilon \delta$ for the ordering $(j \rightarrow m)$ (the weight on the second-leg shock has changed by $J_{mj} \varepsilon$), and symmetrically for $(m \rightarrow j)$; differencing leaves $\varepsilon \delta (J_{mj} - J_{jm})$. From the formula for J , the term $\sum_k u_k \ell_{km} \ell_{kj}$ is symmetric in (j, m) , so $J_{mj} - J_{jm} = \lambda_m^h m_j - \lambda_j^h m_m$. This vanishes for all pairs if and only if $\mathbf{m}$ is proportional to λ^h ; by $\mathbf{m} = \mathbf{L}^\top \mathbf{u}$ and the invertibility of $\mathbf{L}^\top$, that is the same restriction on $\mathbf{u}$ as the zero-wedge condition of Corollary 1. $\square$

Path dependence of the chained index and the failure of the observed Domar weight as a GDP sufficient statistic are, in this economy, one knife-edge apart from each other. Both are manifestations of committed expenditure misaligned with discretionary spending.

How large is it? A quantitative reassurance. On the calibrated sample, I evaluate both sides of (B.1) for the agricultural and manufacturing shocks (10 log points each, adverse, in the two orders), computing the exact chained responses by integrating the closed-form weights along each path. Three facts emerge. First, the local curl $|\lambda_M^h m_A - \lambda_A^h m_M|$ has a sample maximum of 1.56 (Vietnam) and a median of 0.084. Second, the exact reordering effect never exceeds 1.33 log points, even at these deliberately large paired shocks. Third, the closed-form prediction is accurate: the ratio of the exact reordering effect to $\varepsilon\delta$ times the curl is approximately 0.92 across the sample. As in the main text, the result has implications in both directions: chained cross-country accounting is approximately path-independent at observed shock sizes (a genuine reassurance), but the approximation is an empirical fact about magnitudes, not a property of the index. The paper’s two measures, in contrast, require no such reassurance.

The endowment metric over discrete comparisons. Equation (26) is a pointwise statement. Over discrete comparisons, the two relegated measures separate from each other. The chained index integrates the observed weights along the realized path, while the discrete endowment metric of Baqaee and Burstein (2023) integrates Hicksian shares evaluated at final utility, and is endpoint-defined. The paper’s welfare measure needs neither construction: for any two admissible productivity vectors, the exact closed form is

$$\Delta \log C = \sum_{j=1}^n \lambda_j^h \Delta_j + \log \frac{\sigma(A^1)}{\sigma(A^0)},$$

path-independent and valid at any shock size, the discrete counterpart of Theorem 3 used throughout the quantitative work.

C The Second-Order Impacts of Sectoral Shocks

This appendix develops the model’s second-order layer (the curvature of welfare, its global concavity under subsistence, exact formulas for scaled shocks with subsistence thresholds, and the welfare cost of sectoral volatility) and quantifies each piece on the calibrated sample. Fixed-weight GDP needs no second-order theory: by Theorem 2, $\log \text{GDP}$ is globally linear in $\log A$, so this measure of GDP exhibits no amplification, no asymmetry between positive and negative shocks, and no sensitivity to volatility in log units, at any order and for shocks of any size. All curvature in the economy’s aggregates belongs to welfare (and, through the changing weights of Theorem 1, to the chained index of Appendix B).

C.1 The Welfare Hessian

Because relative prices are exactly log-linear in productivities, second-order behavior is governed by one object: the response of the committed shares. Differentiating $u_k = \bar{c}_k q_k$ twice

along a shock direction $d \log A$ gives $d^2 u_k = u_k (d \log q_k)^2$, so that

$$d^2 \sigma = - \sum_{k=1}^n u_k (d \log q_k)^2 = - \sum_{j=1}^n \sum_{m=1}^n H_{jm} d \log A_j d \log A_m, \quad \mathbf{H} := \mathbf{L}^\top \text{diag}(\mathbf{u}) \mathbf{L}. \quad (\text{C.1})$$

Proposition 4 (Second-order impact on welfare). *At every admissible A ,*

$$\frac{\partial^2 \log \mathcal{C}}{\partial \log A_j \partial \log A_m} = - \frac{\sigma H_{jm} + m_j m_m}{\sigma^2}, \quad \text{equivalently} \quad d^2 \log \mathcal{C} = - \frac{1}{\sigma} \sum_{k=1}^n u_k (d \log q_k)^2 - \left(\frac{d\sigma}{\sigma} \right)^2. \quad (\text{C.2})$$

Proof. From $\log \mathcal{C} = \log \text{GDP} + \log \sigma$ and the linearity of $\log \text{GDP}$, $d^2 \log \mathcal{C} = d^2 \log \sigma = d^2 \sigma / \sigma - (d\sigma / \sigma)^2$; insert (C.1) and, for the componentwise form, $d\sigma = \sum_j m_j d \log A_j$ with $H_{jm} = \sum_k u_k \ell_{kj} \ell_{km}$. $\square$

The two terms are two channels of curvature. The first, the base channel, captures the self-reinforcement of committed positions: an adverse shock raises the relative price of committed goods, which raises the committed shares, which makes the next unit of the same shock more costly, at rate $u_k \ell_{kj}^2$ per committed position and scaled by $1/\sigma$. The second, the marginal-utility channel, is the deterioration of the multiplier itself: $d\sigma/\sigma$ squared, exploding as $1/\sigma^2$ near subsistence. In the single-necessity calibration the Hessian element for the agricultural shock collapses to $-u_A (\ell_{AA}/\sigma)^2$, so the ratio of second- to first-order terms for a shock of size δ is at most $|\delta| \ell_{AA} / (2\sigma)$, and approximately that near the subsistence corner, where the committed term dominates the first-order response: half the shock, times the own-multiplier, per unit of discretionary share.

C.2 Global Concavity, Asymmetry, and Subsistence Thresholds

Proposition 5 (Global concavity under subsistence). *If $\bar{c} \geq 0$ (pure subsistence, the baseline configuration), the Hessian (C.2) is negative semidefinite at every admissible A , and $\log \mathcal{C}$ is globally concave in $\log A$ on the admissible region, which is itself convex. Consequently, for any admissible pair $Ae^{+\Delta}, Ae^{-\Delta}$, losses exceed gains:*

$$\underbrace{\log \mathcal{C}(A) - \log \mathcal{C}(Ae^{-\Delta})}_{\text{loss from the adverse shock}} \geq \underbrace{\log \mathcal{C}(Ae^{+\Delta}) - \log \mathcal{C}(A)}_{\text{gain from the favorable shock}}, \quad (\text{C.3})$$

for shocks of any size and in any direction. Fixed-weight GDP satisfies (C.3) with equality, always.

Proof. With $\mathbf{u} \geq 0$, both terms of the directional form in (C.2) are nonpositive for every direction, so the Hessian is negative semidefinite wherever the solution is admissible. Convexity of the admissible region follows because $\sigma(A) = 1 - \sum_k \bar{c}_k e^{-(L \log A)_k}$ is concave in $\log A$ when $\bar{c} \geq 0$ (a sum of concave functions), so $\{\sigma > 0\}$ is convex. With $\bar{c} \geq 0$, interiority is implied by $\sigma > 0$ (Section 2.3), so the admissible region coincides with $\{\sigma > 0\}$. Concavity then gives $\log \mathcal{C}(Ae^{+\Delta}) + \log \mathcal{C}(Ae^{-\Delta}) \leq 2 \log \mathcal{C}(A)$, which rearranges to (C.3). The GDP statement is linearity (Theorem 2). $\square$

Exact responses to scaled shocks and the subsistence threshold. Scale a fixed shock direction Δ by $t \geq 0$, so that $\log A(t) = \log A^0 + t\Delta$, and define $y_k = -(\mathbf{L}\Delta)_k$ (positive for adverse food-chain shocks). Committed shares then evolve exactly as $u_k(t) = u_k^0 e^{ty_k}$, and

$$\Delta \log \mathcal{C}(t) = t \sum_{j=1}^n \lambda_j^h \Delta_j + \log \frac{1 - \sum_k u_k^0 e^{ty_k}}{1 - \sum_k u_k^0}, \quad (\text{C.4})$$

valid up to the subsistence threshold t^* : the first $t > 0$, if one exists, at which $\sum_k u_k^0 e^{ty_k} = 1$ (on the adverse single-necessity scalings used below, the committed bill is monotone and the threshold unique): there the committed bundle exhausts income, the welfare level collapses to zero ($\log \mathcal{C} \rightarrow -\infty$), and the interior equilibrium ceases to exist. For the single-necessity calibration and the adverse own shock, the threshold takes the closed form

$$|\Delta^*| = \frac{|\log u_A|}{\ell_{AA}} :$$

the distance to the wall is the (log) headroom in the committed share, divided by the network's own-multiplier: demand determines the wall's position, the network the speed of approach. Along the same direction, for adverse-leg sizes $0 < \Delta < |\Delta^*|$, adding (C.4) at $\pm\Delta$ gives the exact loss-gain asymmetry in closed form,

$$\text{AS}(\Delta) := |\Delta \log \mathcal{C}(-\Delta)| - |\Delta \log \mathcal{C}(+\Delta)| = -\log \frac{1 - 2u_A \cosh(\ell_{AA}\Delta) + u_A^2}{(1 - u_A)^2} \geq 0,$$

which is increasing in both the committed share and the own-multiplier, and diverges as the adverse leg approaches the wall.

C.3 The Welfare Cost of Sectoral Volatility

Corollary 3 (The welfare price of volatility). *Let ε be a mean-zero vector of sectoral shocks, admissible on its support, with covariance matrix Σ and third moments $\mathbb{E}\|\varepsilon\|^3 = o(\|\Sigma\|)$, and let $\mathbf{y} = \mathbf{L}\varepsilon$ be the induced log-price movements (up to sign). Then*

$$\mathbb{E}[\Delta \log \mathcal{C}] = -\frac{1}{2\sigma} \sum_{k=1}^n u_k \text{Var}(y_k) - \frac{1}{2\sigma^2} \text{Var}\left(\sum_{k=1}^n u_k y_k\right) + o(\|\Sigma\|), \quad (\text{C.5})$$

while $\mathbb{E}[\log \text{GDP}] = \log \text{GDP}$ exactly.

Proof. Take expectations of the second-order expansion: the first-order term vanishes by mean zero, and the quadratic term is the expectation of (C.2)'s directional form, with $\mathbb{E}[(d \log q_k)^2] = \text{Var}(y_k)$ and $\mathbb{E}[(d\sigma)^2] = \text{Var}(\sum_k u_k y_k)$. The GDP statement is exact linearity. $\square$

The two loadings mirror the base and marginal-utility channels: the first loads on *basket-price risk* (the variance of each committed good's price, weighted by the committed position), and the second on *subsistence risk*, the variance of the committed bill itself, divided by σ^2 . Both load on the network *squared* (through $\mathbf{y} = \mathbf{L}\varepsilon$ and $\mathbf{H}$): volatility pricing is where the network-subsistence interaction is at its sharpest.

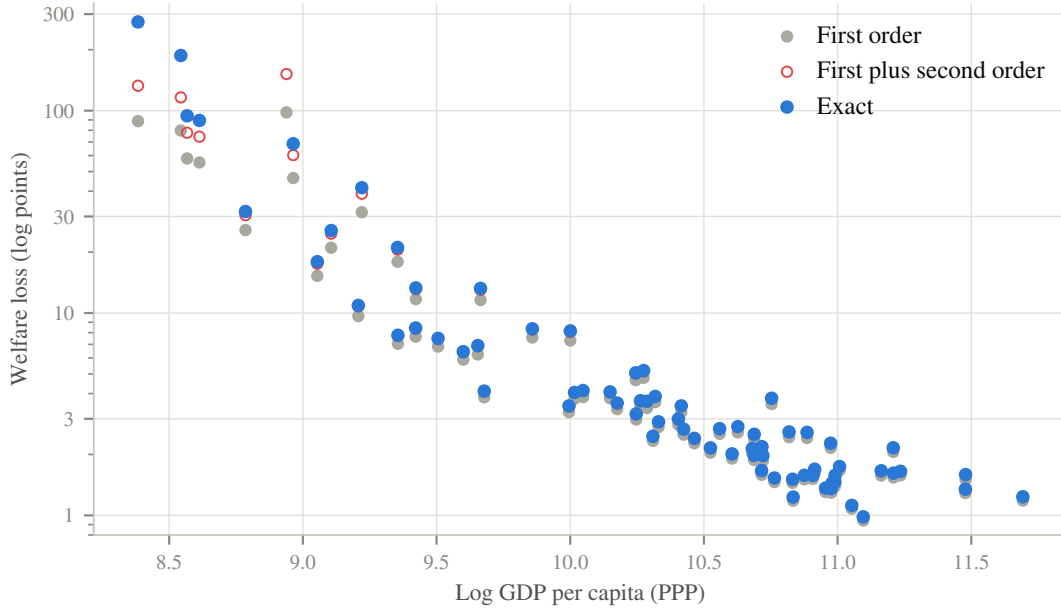


Figure C.1: The welfare response by order of approximation: first order, first plus second, and exact

Notes: Absolute welfare losses from the 10 percent agricultural shock for included economies ($n = 74$), in log points: the first-order response (gray), the first-order plus second-order response, from the Hessian term (C.2) (red open), and the exact response (C.4) (blue). The horizontal axis is the log of GDP per capita; the vertical axis is logarithmically spaced.

Remark 1 (Cardinalization). Corollary 3 expresses the cost of volatility in expected log-welfare units; equivalently, the certainty-equivalent loss when ex-ante preferences over $\mathcal{C}$ are logarithmic, $v(\mathcal{C}) = \log \mathcal{C}$. Under CRRA ex-ante curvature $\gamma \neq 1$, the certainty-equivalent loss adds a risk premium $\frac{\gamma-1}{2} \text{Var}(\Delta \log \mathcal{C})$ to (C.5). To leading order, the premium loads on the variance of the first-order exposure $\sum_j (\lambda_j / \sigma) \varepsilon_j$, a different loading than those of (C.5), so the cross-country gradient and composition of the certainty-equivalent price can shift with γ .

C.4 Quantification

How much does second order add to the shock experiments? Figure C.1 compares, for the 10 percent agricultural shock, the first-order response, the first-order plus second-order response, and the exact response (C.4), economy by economy. The per-country ordering $|\text{first}| \leq |\text{first+second}| \leq |\text{exact}|$ holds without exception, as follows from the exact response (C.4): on the adverse leg, every term beyond the first order adds to the loss. The correction is a rounding error among the richest economies (second-to-first ratio 0.044 at the richest-quartile median) and material only near the corner (0.21 at the poorest-quartile median). Beyond the quadratic, the remainder is negligible away from the wall (medians 0.0024 and 0.041 of the exact response in the richest and poorest quartiles).

The welfare price of volatility. Figure C.2 evaluates equation (C.5) at common sectoral shock moments (agricultural total factor productivity volatility from long international agricultural-productivity panels, and manufacturing and services volatility from international sectoral labor-productivity panels, pooled across economies over a fixed 1990–2019 protocol), so that, as with the shock experiments, cross-country differences reflect exposure rather than shock size.²⁷ The gradient is the steepest in the paper: the median expected annual welfare change is -2.3×10^{-4} natural-log units in the richest income quartile and -205×10^{-4} in the poorest, a factor of 89, an order of magnitude steeper than the gradient of the multiplier itself: the price is second order, and its single-necessity loading is the committed exposure squared per unit of committed share, $[(u_A/\sigma)\ell_{AA}]^2/u_A = u_A(\ell_{AA}/\sigma)^2$. The composition of the price also rotates along the development path: the subsistence-risk share of the price is 0.06 at the richest-quartile median and 0.61 at the poorest. The quadratic formula is validated against exact evaluations on two-point shock distributions (sample-median accuracy ratio 0.999; worst case 0.82); the margin and shock-scale variants of Appendix F dampen the price at the poorest-quartile median by factors of 0.55 and 0.69.

D An Example Combining a Necessity and a Luxury: Agriculture, Manufacturing, Services

This appendix works out the three-sector example mentioned in Section 3.3, in which both signs of the committed position coexist, so that the necessity mechanics of Example 2 and the luxury mechanics of Example 3 compete within one economy.

Example 5 (Agriculture, manufacturing, services). Let $n = 3$, with agriculture (A) a necessity, services (S) a luxury, and manufacturing (M) neutral: $\bar{c}_A > 0$, $\bar{c}_M = 0$, $\bar{c}_S < 0$. Then

$$\sigma = 1 - u_A - u_S = 1 - \bar{c}_A q_A + |\bar{c}_S| q_S, \quad m_j = \bar{c}_A q_A \ell_{Aj} - |\bar{c}_S| q_S \ell_{Sj} \gtrless 0,$$

and, for a shock to agriculture:

(i) **Expenditure shares.**

$$d\tilde{\beta}_A = [-(1 - \beta_A)u_A \ell_{AA} - \beta_A |u_S| \ell_{SA}] d\log A_A, \quad (D.1)$$

$$d\tilde{\beta}_S = [(1 - \beta_S)|u_S| \ell_{SA} + \beta_S u_A \ell_{AA}] d\log A_A, \quad (D.2)$$

$$d\tilde{\beta}_M = \beta_M m_A d\log A_A = \beta_M (u_A \ell_{AA} - |u_S| \ell_{SA}) d\log A_A. \quad (D.3)$$

²⁷Sectoral moments: agricultural TFP growth from the USDA Economic Research Service's International Agricultural Productivity data product; manufacturing and services labor-productivity growth from the GGDC/UNU-WIDER Economic Transformation Database. Because the moments are common across economies, rescaling them uniformly moves the level of every price in Figure C.2 but leaves cross-country ratios unchanged. The shock-scale variant in Appendix F, which scales each economy's agricultural moment by its primary-agriculture weight, bounds the level.

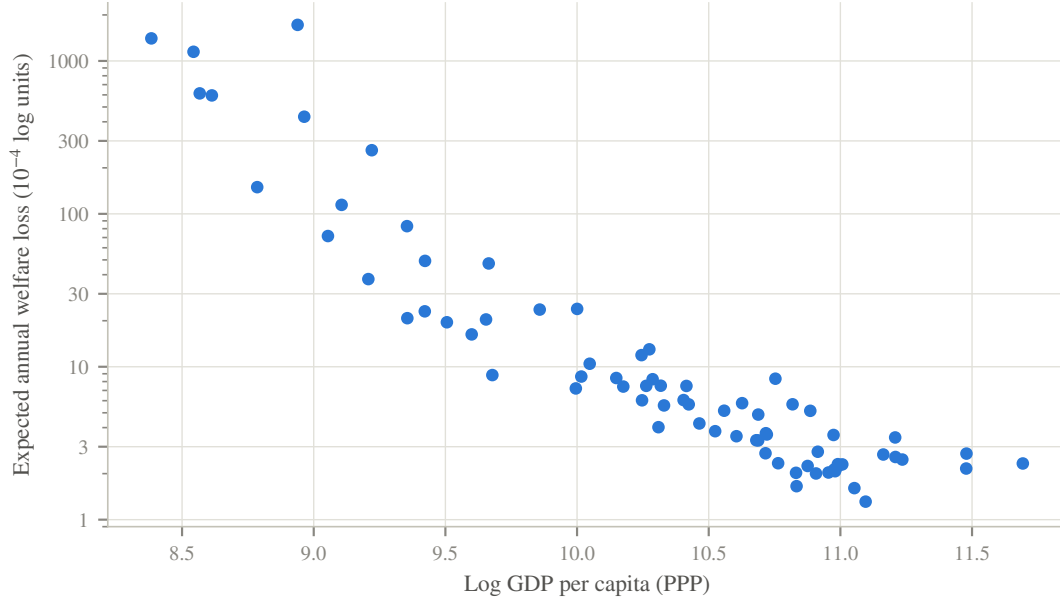


Figure C.2: The welfare price of sectoral volatility

Notes: Expected annual welfare change (C.5) at common sectoral shock moments, per included economy ($n = 74$), in absolute value. The horizontal axis is the log of GDP per capita; the vertical axis is in units of 10^{-4} natural logs and is logarithmically spaced.

The first bracketed coefficient is negative and the second positive, while manufacturing's carries the sign of the net income effect m_A : for a positive agricultural shock ($d \log A_A > 0$), agriculture's share falls, services' rises, and manufacturing's can move either way.

(ii) **Domar weights.** For each industry $i \in \{A, M, S\}$,

$$d\lambda_i = \left[u_A \ell_{AA} (\lambda_i^h - \ell_{Ai}) - |u_S| \ell_{SA} (\lambda_i^h - \ell_{Si}) \right] d \log A_A. \quad (\text{D.4})$$

(iii) **Aggregates.**

$$d \log \text{GDP} = \lambda_A^h d \log A_A, \quad d \log C = \frac{\lambda_A}{\sigma} d \log A_A = \left[\lambda_A^h + \frac{u_A \ell_{AA} - |u_S| \ell_{SA}}{\sigma} \right] d \log A_A.$$

Two income effects. A positive agricultural shock enriches the household through the necessity channel: cheaper food frees committed budget, the term $u_A \ell_{AA}$, always active since $\ell_{AA} \geq 1$. To the extent that agriculture is upstream of services ($\ell_{SA} > 0$), it also impoverishes the household through the luxury channel: cheaper services devalue the home endowment, the term $-|u_S| \ell_{SA}$. At the extremes, the two channels push the composition of demand the same way. Food's share falls twice over in (D.1): cheaper food shrinks the committed tier of the food share, and the devalued home endowment shrinks discretionary income, scaling down the discretionary tier as well. Services' share rises twice over in (D.2): the negative committed position shrinks in magnitude as services cheapen, and part of the food-relief gain is spent

on services. Structural change from agriculture toward services after agricultural productivity gains is therefore unambiguous. Only the middle sector inherits the net income effect m_A , rising if and only if subsistence relief dominates endowment devaluation, by (D.3).

Sales. Equation (D.4) nets the two channels industry by industry. For agriculture itself, the first term is negative (Lemma 1: $\lambda_A^h < \ell_{AA}$), and the second is negative as well whenever services rely on agriculture less than the preference-weighted bundle does ($\ell_{SA} < \lambda_A^h$), the empirically typical configuration, so λ_A falls with agricultural productivity and rises as the economy is impoverished, by more the poorer the economy. Sign is genuinely conditional here. With two committed positions of opposite sign, network detail decides, which is why the unconditional results of the main text are stated for single-position configurations.

Rich versus poor. Consider two economies with identical $(\Omega, \beta, \bar{c})$ hit by the same adverse agricultural shock, one poor (an adverse food-productivity history, hence a high q_A and a large committed food share) and one rich (a low q_A and, as an empirically motivated configuration rather than an implication of richness, a sizable home-services value $|\bar{c}_S|q_S$). Fixed-weight GDP responds identically in the two economies: λ_A^h contains neither $\bar{c}$ nor A . Chained GDP can respond more in the poor economy: its coefficient is the observed λ_A , which is increasing in the committed food share, though the rich economy's home-services position also moves λ_A , so the ranking requires the food-share difference to dominate. Welfare responds far more in the poor economy: $\sigma < 1$ there amplifies every shock by $1/\sigma$, while the rich economy may well have $\sigma > 1$ when home production dominates, damping its welfare response below even its GDP response. The gap between the two economies' responses to the same shock is governed by the single statistic λ_A/σ . This is the model-consistent statement of the paper's motivating fact, together with an exact account of which measures carry the asymmetry (welfare and chained GDP, not fixed-weight GDP) and of the parameters through which it should be quantified.

E Data and Calibration: Details

Input-output tables. The source is the OECD Inter-Country Input–Output (ICIO) database, 2025 edition, whose methodology is documented in Yamano, Webb, Cimper, Pazos, and Kohli (2023). I use the 2017 reference-year table in the baseline (and 2021 for the robustness check). The table covers 80 economies individually plus a rest-of-world aggregate, at 50 industries, in current US dollars at basic prices. For each economy, I aggregate flows to the three sectors of Section 4.1: A (ISIC A01–A03 and C10–C12), M (B–F excluding C10–C12), and S (the remainder), and then take coefficients. Following Herrendorf et al. (2013), the baseline uses total cost shares, summing each buyer's purchases over all source countries, so that imported inputs are treated as if they were domestically produced with the same cost structure. A domestic-flows-only variant, which drops this treatment, is a robustness row in Appendix F. The residual primary-factor share is value added at basic prices plus taxes less subsidies on products levied on the industry's intermediate purchases (the ICIO's TLS row), the single-composite-factor

treatment, under which $\alpha_i + \sum_j a_{ij} = 1$ holds exactly in the data. “Labor share” throughout the model is this composite value-added share. Final-demand shares $\tilde{\beta}$ use household-facing absorption (household and non-profit consumption, government consumption, and residents’ direct purchases abroad), the consumption concept of the structural change literature; exports never enter (absorption closure), and a domestic-absorption variant including investment is computed alongside for robustness. One caveat regarding pricing is that ICIO flows are at basic prices while the anchor and the ICP data are at purchaser prices. The unrouted purchaser margin is the one remaining placement approximation, and the margins variant in Appendix F bounds it (it works against the paper’s asymmetries).

Prices, expenditures, and incomes. Food price levels are from the International Comparison Program, 2017 cycle (World Bank, 2020): the ICP category is food and non-alcoholic beverages, and price levels are expressed relative to the US. The consumption budgets E^c are not ICP objects: they are the household-facing absorption of the ICIO final-demand accounts (the same vector behind $\tilde{\beta}$ and λ), in current US dollars at market exchange rates (the same basis as the anchored committed value), divided by World Bank population. Incomes are PPP GDP per capita from the Penn World Table 11.0 (expenditure-side real GDP at chained PPPs over population, in 2021 reference dollars; the 2017 observation year supplies the income axis) (Feenstra et al., 2015; Groningen Growth and Development Centre, 2025). The anchor series is the US national accounts’ share of food and beverages purchased for off-premises consumption in personal consumption expenditures (NIPA), annually over 1947–2010. The NIPA category includes alcoholic beverages purchased off premises, while the ICP food category excludes them, so the alcohol-inclusive ICP variant in Appendix F is the category-matched robustness row (the integrated sector itself additionally contains forestry, fishing, and tobacco).

The anchor regression, mechanically. Equation (27) is estimated on the anchor series as $s_t = a + br_t$ with s_t the NIPA food share and $r_t = p_{A,t}/E_t$, which is linear in (a, b) . The map $(a, b) \mapsto (\hat{\omega}_A, \bar{c}_A^*) = (a, b/(1-a))$ is a smooth bijection on $a < 1$ (the estimate is interior, so the constrained and unconstrained problems coincide), and ordinary least squares on the linear form delivers the exact nonlinear least-squares estimate of (27) under share-error loss. Rebas-ing the food price index to one in 2017 denominates $\bar{c}_A^*$ directly in 2017 US dollars per person per year, the unit in which equation (28) uses it. Because the regression’s residuals are strongly persistent, all standard errors are serial-correlation robust, and the block bootstrap resamples residual blocks with the regressor path held fixed. The loss function itself is a specification choice, and re-estimating in dollar-loss (level) form or under a Prais–Winsten AR(1) error model (the variant that retains the first observation, which carries the low-frequency identifying variation; quasi-differencing estimators that drop it behave like the attenuated first-difference form) spans the specification band reported with the estimates in Section 4.2, alongside the end-year band and the separately-reported start-year scenario.

Time-series diagnostics. The anchor regression runs in levels on six decades of trending series,

and the standard diagnostics are reported here, with the caveat that sixty-four annual observations of a share bounded between zero and one fit no unit-root null exactly. KPSS tests reject level stationarity for both series (statistics 1.16 for the share and 1.13 for the price-to-expenditure ratio, against a five-percent critical value of 0.46; $p < 0.01$); augmented Dickey–Fuller tests fail to reject a unit root in the share ($p = 0.09$ with a constant, 0.98 with a trend), while the verdict for the ratio is specification-dependent ($p = 0.004$ with a constant, at a boundary lag length, but 0.88 with a trend). An Engle–Granger test does not reject the absence of cointegration between the two ($t = -2.10$, $p = 0.47$): the level relation is not statistically certified as cointegrating, and the reader should know it, though tests of this family have little power against six decades of strongly persistent residuals, and equation (27) is the model’s own share identity rather than an atheoretic regression. The operative check is therefore specification rather than asymptotics: re-estimating the slope in first differences (which discards the low-frequency variation that carries the identification, the attenuation noted above for quasi-differencing estimators) yields $\bar{c}_A^* = \$1,618$ (−12.4 percent), and the long run of an unrestricted error-correction model (ARDL(1,1), adjustment coefficient −0.045 per year) yields \$1,613 (−12.7 percent); both sit essentially at the borrowed-anchor level (\$1,639.90), whose full-pipeline consequences the robustness grid of Appendix F reports as its own row. The anchor-level uncertainty that matters for the paper’s conclusions is thus the one already carried throughout: the specification band, the separately-labeled start-year scenario, and the calibration bounds of Appendix F.

Country demand states. Given the anchored committed value $\bar{c}_A^*$, each economy’s committed share follows from (28), and the preference shares from the share equation (8) evaluated at the US under the two zero restrictions: $\beta_A = (\tilde{\beta}_A^{\text{US}} - u_A^{\text{US}}) / \sigma^{\text{US}}$, $\beta_M = \tilde{\beta}_M^{\text{US}} / \sigma^{\text{US}}$, and $\beta_S = \tilde{\beta}_S^{\text{US}} / \sigma^{\text{US}}$, which sum to one by the US share identity. The model’s structure supplies a second, share-anchored reading that uses no food-price data: under the zero restrictions, the services share equation implies an independent estimate of each economy’s discretionary share, $\sigma_S^c = \tilde{\beta}_S^c / \beta_S$. Section 4.2 reports what it does and does not validate (agreement in how committedness moves with GDP per capita; essentially no correlation at a given level of GDP per capita; a disagreeing manufacturing facet), and Appendix F promotes it to a calibration bound. The admissibility audit then evaluates Assumption 2 economy by economy ($u_A^c < 1$, interior consumption); the failures are the affordability census of Appendix F.